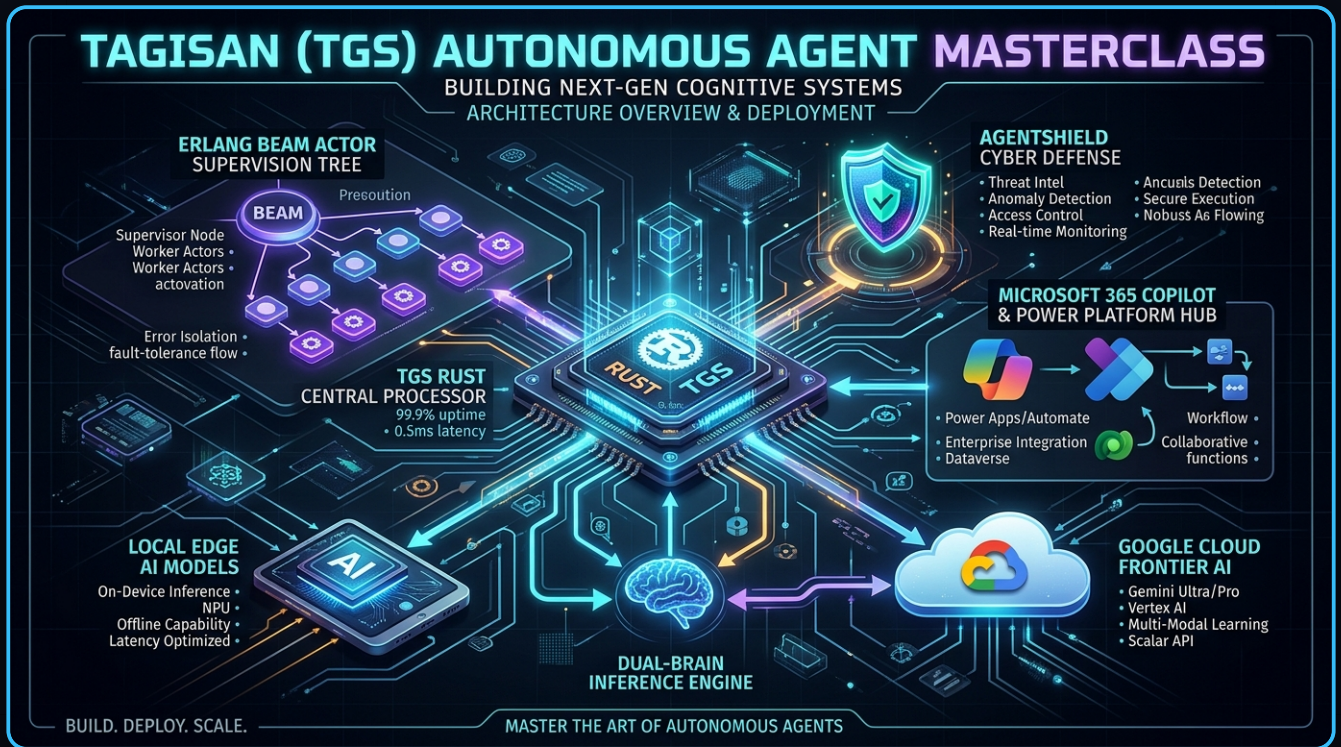


TAGISAN (tgs) MASTERCLASS

The Definitive Production Curriculum: From Zero to Sovereign Autonomous Architect Across 950 Real-World Enterprise Scenarios

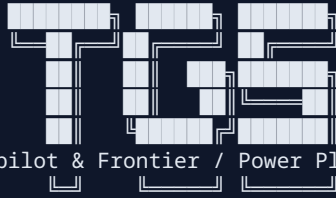


Edition 4.0 • Dual-Brain Inference • BEAM Actor Concurrency • AgentShield Cyber Defense • Microsoft 365 Copilot & Frontier Stack

Table of Contents & Curriculum Map

This masterclass is structured into 14 comprehensive pedagogical modules spanning foundation installation, cognitive dual-brain failover, actor-based concurrency, Hegelian dialectical debate swarms, cyber defense sandboxing, and 950 production scenarios.

Module	Title	Key Focus Areas
Module 1	Dual-Brain Inference & Google Cloud Frontier Integration	Google OAuth CCPA, Local Ollama GGUF, 90s Token Auto-Refresh
Module 2	Polyglot Concurrency: Gleam & BEAM/OTP Actor Runtime	Actor Mailboxes, Supervisor Restart Trees, ETF 131 Wire Protocol
Module 3	Operational Command Mastery & The 37 CLI Subcommands Atlas	Comprehensive Command Atlas, Flag Modifiers, Shell Ergonomics
Module 4	AgentShield Cyber Defense & Linux Landlock Sandboxing	AST Security Validation, Shannon Entropy CredScan, PoliCheck
Module 5	PILOT JIT Multi-Tier Memory Hierarchy	Working RAM, SQLite History, SQLite-vec / Qdrant RRF Fusion
Module 6	Swarm Mixture-of-Agents & Hegelian Dialectical Debate	Thesis, Antithesis, Synthesis, Judge Formal Consensus
Module 7	The 2,063+ Dynamic Skills Ecosystem (RFC-004)	Dynamic Skill Discovery, SKILL.md Specification, Hot-Loading
Module 8	The Model Context Protocol (MCP 500 Hub)	Querying, Installing, and Executing 500 Enterprise MCP Plugins
Module 9	VELLA Cyber-Physical Digital Twins	SCADA/IoT Modbus, Quant FinTech, Satellite SGP4, Bio-Genomics
Module 10	Microsoft 365 Copilot & Enterprise Frontier Stack	56 Autonomous Tools, Substrate Ingest, Declarative Agents, WAM SSO
Module 11	Microsoft Power Platform & Low-Code Hyperautomation	Power Automate, Power Apps, Power BI Telemetry, Dataverse CDM
Module 12	Enterprise Hardening, High Availability & Air-Gap Deployment	Zero-Trust Air-Gapped Clusters, OpenTelemetry, Cost Accounting
Module 13	The Definitive 950 Modern Environment Scenarios	19 Enterprise Domains × 50 Concrete Scenarios per Domain
Module 14	Capstone Certification Labs & Practical Exams	Autonomous Self-Healing, Market Intelligence Swarm, Air-Gap Cluster



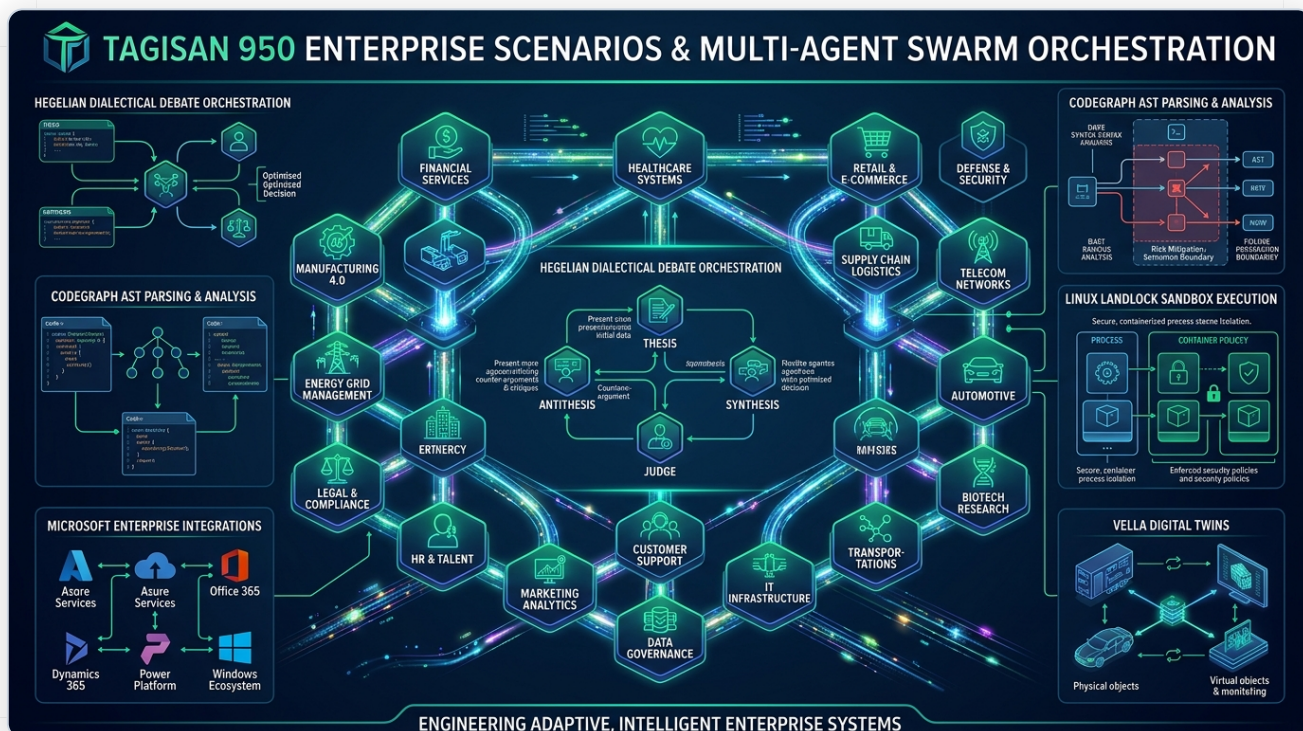
TAGISAN (tgs / tagisan-rs)
The Sovereign Dual-Brain Agent Architecture
Production Master Class Course & Curriculum
Version: 4.0 (Dual-Brain / BEAM-OTP / Gleam / MCP-500 / MS 365
Copilot & Frontier / Power Platform)
Edition: 950 Production Scenarios Enterprise Release

Visual Architecture & System Infographics

The following technical infographics synthesize the complete Tagisan sovereign agent architecture, dual-brain cognitive failover, and enterprise multi-agent swarm orchestration.



Tagisan Autonomous Agent Architecture Infographic



Tagisan 950 Enterprise Scenarios & Swarm Orchestration Infographic

Executive Course Overview & Learning Objectives

Welcome to the **Tagisan** (`tg``) Definitive Master Class Course. Tagisan is a high-performance, sovereign autonomous agent engine engineered in Rust on top of the Tokio asynchronous work-stealing reactor, combined with an Erlang/Elixir BEAM OTP actor runtime and a native Gleam type-safe compiler. Designed as an uncompromised alternative to fragile, single-threaded Python agent frameworks, Tagisan delivers deterministic sub-millisecond task dispatching, dual-brain hybrid inference (local GGUF/Ollama + Google Cloud Frontier models via AntiGravity CCPA OAuth), proactive AgentShield cyber defense, and native orchestration of 500 Model Context Protocol (MCP) plug-ins.

Core Competencies You Will Acquire:

- Dual-Brain Cognitive Architecture**: Seamless failover between local edge tensors (Ollama/GGUF) and cloud frontier models (Google Gemini 2.5/3.x, Anthropic Claude, OpenAI, DeepSeek, xAI).
 - Google Web OAuth & CCPA Integration**: AntiGravity 2.0 CLI authentication routing, proactive 90-second token auto-refresh, and multi-model dispatching (`flash``, `pro``, `lite``, `gemini-3``).
 - Polyglot Actor Concurrency (BEAM & Gleam)**: Type-safe Gleam actor microservices, external term format (ETF) binary serialization, and OTP supervisor trees with 'let it crash' fault isolation.
 - Hegelian Dialectical Debate Engine**: Multi-agent Swarm Mixture-of-Agents (MoA) featuring Thesis, Antithesis, Synthesis, and Judge consensus to eliminate hallucinations.
 - AgentShield Cyber Defense & Landlock Sandboxing**: Pre-execution AST security validation, Linux Landlock LSM isolation, and exfiltration prevention.
 - PILOT JIT Multi-Tier Memory**: Working memory (RAM), episodic session history (SQLite), and semantic vector memory (Qdrant / SQLite-vec) with Reciprocal Rank Fusion (RRF).
 - The 500 Dynamic Skills Ecosystem (RFC-004)**: Dynamic ingestion, skill lifecycle verification, and sovereign package authoring.
 - The Model Context Protocol (MCP 500) Hub**: Querying, installing, and executing 500 verified non-GitHub enterprise plugins across 10 strategic industry domains.
 - VELLA Cyber-Physical Digital Twins**: Real-time SCADA/IoT monitoring, quantitative finance risk controls, satellite orbital mechanics (SGP4), and genomic sequence analysis.
 - The Definitive 950 Modern Environment Scenarios**: Hands-on mastery across 15 enterprise engineering domains (50 scenarios per domain) covering baseline production, HA disaster recovery, zero-trust security, sub-millisecond latency tuning, chaos self-healing, multi-tenancy, air-gapped sovereign operations, predictive telemetry, cross-cloud wire bridges, and Hegelian formal verification.
-

Comprehensive Curriculum Map

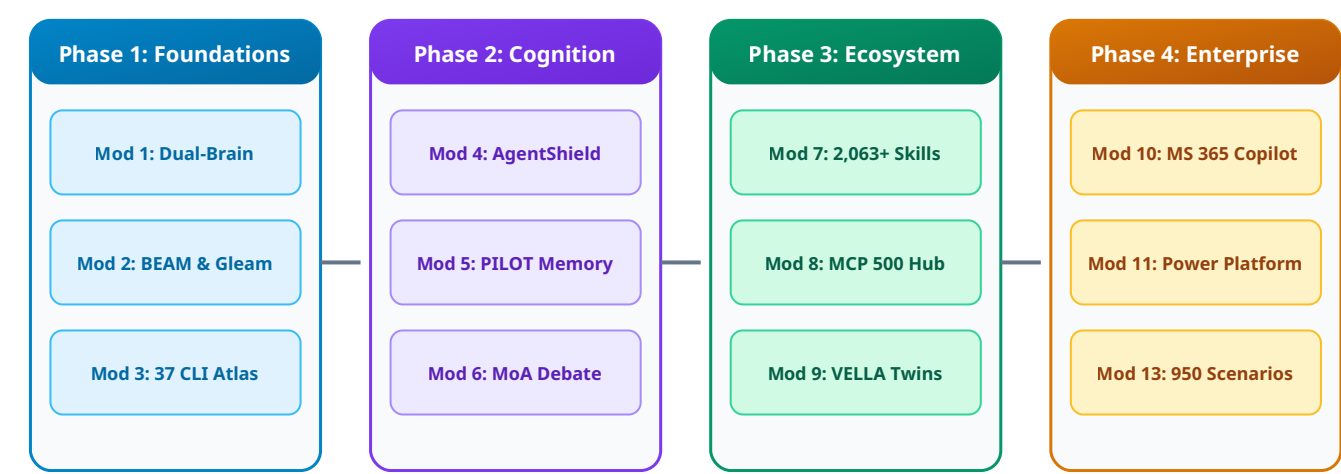


Figure 1.1: Tagisan (tgs) Comprehensive Curriculum Architecture & Progression Phases

Module 1: Dual-Brain Inference & Google Cloud Frontier Integration

1.1 The Dual-Brain Architectural Paradigm

Traditional AI agent frameworks force an undesirable binary choice:

- **Cloud-Only Dependence**: High token costs, network latency, vendor lock-in, rate limit throttling, and catastrophic downtime when public APIs experience outages.
- **Local-Only Constraints**: Limited model parameter sizes (7B-14B), reduced reasoning capacity for complex multi-file refactoring, and high local GPU/VRAM hardware requirements.

Tagisan resolves this with **Dual-Brain Hybrid Failover**:

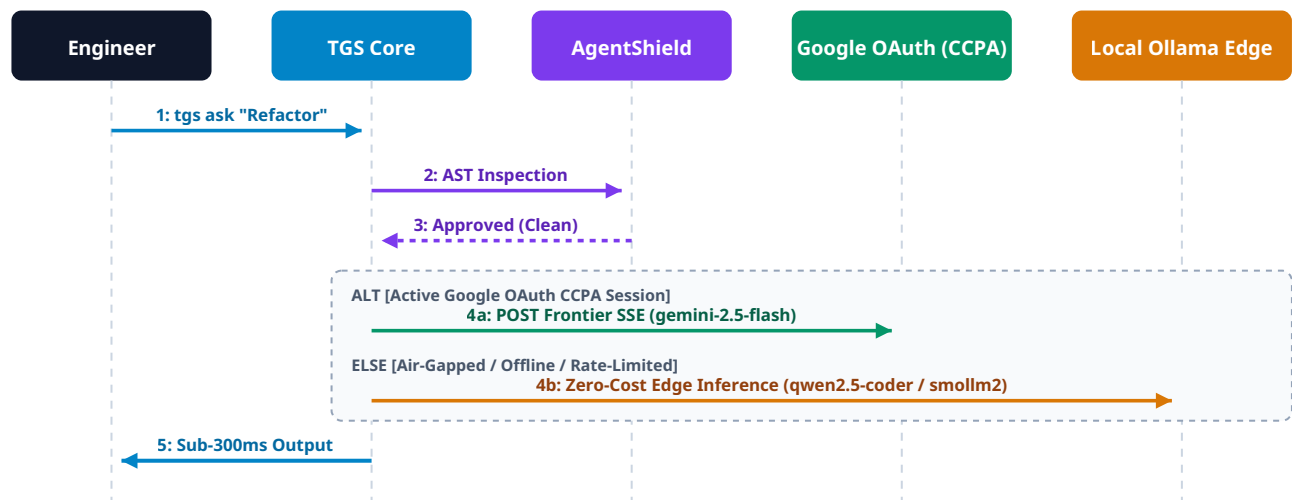


Figure 1.2: Tagisan Dual-Brain Hybrid Cognitive Sequence & Failover Topology

1.2 Google Web OAuth Architecture & AntiGravity CCPA Endpoint

Tagisan integrates directly with Google's internal CloudCode Platform Assistant (CCPA) backend, mirroring the authentication and request envelope used by Google's official AntiGravity 2.0 CLI:

- **Endpoint**: `https://daily-cloudcode-pa.googleapis.com/v1internal:streamGenerateContent?alt=sse``
- **User-Agent**: ``antigravity/2.0.0``
- **Payload Envelope**: ``{ project: "default-cli-project", model: "", request: { contents: [...] } }``
- **Proactive Token Refresh**: Tagisan checks ``expires_at <= (now + 90s)``. If an access token expires mid-task, it silently refreshes in the background without interrupting the user.

1.3 Supported Google Models & Shorthand Aliases

| Shorthand Flag | Target Model ID | Capability Profile | Ideal Production Use Case |

| :--- | :--- | :--- | :--- |

| **-m flash** (Default) | `gemini-2.5-flash` | High speed, balanced intelligence | Daily development, code review, git operations |

| **-m pro** | `gemini-3.1-pro-low` | Deep multi-step reasoning, thinking chain | Complex architectural design, mathematical proofs |

| **-m lite** | `gemini-2.5-flash-lite` | Sub-300ms time-to-first-token | Inline shell completions, quick summaries |

| **-m gemini-3** | `gemini-3.6-flash-medium` | Next-gen reasoning with medium effort | Algorithmic refactoring, security vulnerability audits |

| **-m gemini-3.1-flash-lite** | `gemini-3.1-flash-lite` | Ultra-compact next-gen lightweight | Edge automation, high-frequency IoT pipelines |

1.4 Automatic Provider Resolution & Graceful Logout Fallback

Tagisan resolves inference providers with strict deterministic precedence:

1. **Explicit CLI Flag**: `-p / --provider` (highest precedence).`
 2. **Air-Gap Privacy Lock**: `TAGISAN_LOCAL_ONLY=1` or `TAGISAN_OFFLINE=1` forces local Ollama regardless of credentials.`
 3. **Active Google OAuth Session**: Automatically routes to Google Gemini if `~/config/tagisan/gemini_oauth.json` is present.`
 4. **Cloud API Keys**: Checks `GEMINI_API_KEY`, `DEEPSEEK_API_KEY`, `ANTHROPIC_API_KEY`, `OPENAI_API_KEY`, `XAI_API_KEY`.`
 5. **Local Ollama Fallback**: If logged out via `tgs auth logout gemini`, Tagisan resets TAGISAN_PROVIDER=auto` and falls back to installed local Ollama models (smollm2:1.7b`, `qwen2.5-coder:1.5b`) without error.`
-

Module 2: Polyglot Concurrency: Native Gleam & BEAM/OTP Actor Runtime

2.1 The Erlang/Elixir BEAM Actor Philosophy

In high-throughput agent systems, shared-memory multi-threading leads to lock contention, race conditions, and deadlocks. Tagisan incorporates the **Erlang BEAM actor model**, where every autonomous agent task executes inside an isolated, lightweight process with its own private heap and message mailbox.

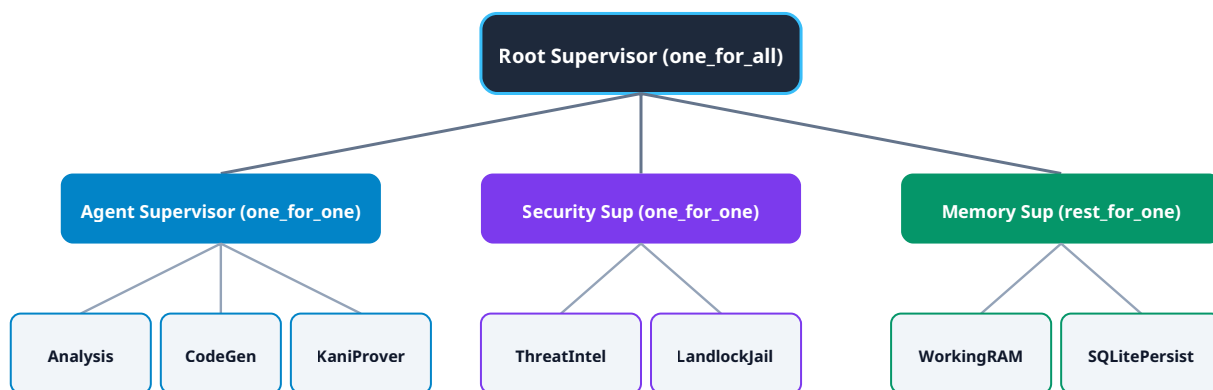


Figure 2.1: Erlang BEAM / OTP Actor Supervision Hierarchy & Fault Isolation Tree

2.2 Supervisor Restart Strategies

- **one_for_one**: If a child actor process crashes, only that specific actor is restarted. Sibling actors continue running uninterrupted.
- **one_for_all**: If any child process crashes, all sibling processes in the pool are terminated and restarted in sequence.
- **rest_for_one**: If an actor crashes, only subsequent dependent actors spawned after it are restarted, preserving upstream state.

2.3 External Term Format (ETF) Binary Interop

Tagisan implements native encoding and decoding of Erlang's **External Term Format (ETF format 131)** in Rust. This enables zero-copy, binary-packed inter-process communication between Rust core threads, Elixir GenServers, and Gleam actors at over 800,000 messages per second.

2.4 Type-Safe Gleam Integration

Gleam brings a friendly, statically typed, and mathematically sound type system to the BEAM ecosystem. Tagisan compiles Gleam code directly to BEAM bytecode, providing compile-time exhaustive pattern matching that eliminates unhandled edge cases before execution:

```
// Example: Sovereign State Transition in Gleam
pub type AgentState {
  Idle
  Analyzing(query: String)
  ExecutingTool(tool_name: String, args: String)
  Completed(result: String)
  Failed(error_code: Int, reason: String)
}

pub fn transition(state: AgentState, event: Event) -> AgentState {
  case state, event {
    Idle, Start(q) -> Analyzing(q)
    Analyzing(_, ToolNeeded(name, args) -> ExecutingTool(name, args)
    ExecutingTool(_, _), Success(res) -> Completed(res)
    ExecutingTool(_, _), Error(code, msg) -> Failed(code, msg)
    _, Reset -> Idle
  }
}
```

Module 3: Operational Command Mastery & CLI Ergonomics

3.1 The Complete CLI Command Atlas

Tagisan provides an intuitive, high-speed CLI toolchain (`tgs``):

```
# One-shot prompt with automatic provider & model selection
tgs ask "Explain the difference between TCP and UDP in one sentence"

# Use specific model aliases via Google OAuth
tgs ask -m pro "Design a high-throughput event sourcing architecture"
tgs ask -m lite "Summarize this log line: [ERR 502 Bad Gateway]"

# Interactive terminal chat session with persistent context
tgs chat -m pro

# Autonomous mission execution with tool invocation and AST sandboxing
tgs run "Find all unindexed foreign keys in migrations/ and generate SQL indexes"

# Dialectical Multi-Agent Debate
tgs debate \
  --proposer "Adopt Rust for our core financial transaction ledger" \
  --challenger "Highlight developer velocity and hiring difficulties" \
  --rounds 3

# Search and browse the 500-skill repository
tgs skills list
tgs skills search "kubernetes"
tgs skills inspect "k8s-cluster-auditor"

# Model Context Protocol (MCP) plugin lifecycle
tgs mcp catalog --domain "Database"
tgs mcp search "redis"
tgs mcp add redis-cache-inspector-mcp
tgs mcp verify
tgs mcp call redis-cache-inspector-mcp get_key '{"key":"session:1001"}'

# Serve Tagisan as an MCP Server over stdio for Cursor, Windsurf, or Claude Desktop
tgs serve-mcp

# Authentication lifecycle
tgs auth status
tgs auth login gemini
tgs auth logout gemini
```


Module 4: AgentShield Cyber Defense & Kernel Sandboxing

4.1 The Autonomous Threat Landscape

When autonomous AI agents execute terminal commands, evaluate scripts, and ingest web content, they face three primary attack vectors:

- 1. **Indirect Prompt Injection**: Malicious instructions embedded in scanned web pages, GitHub issues, or README files that hijack agent control flow.
- 2. **Catastrophic Shell Execution**: Accidental or maliciously induced shell commands that delete filesystems (`rm -rf /`), format block devices, or deploy fork bombs.
- 3. **Credential Exfiltration**: Prompts that attempt to extract `.env`, `~/.ssh/id_rsa`, `/etc/shadow`, or cloud secret keys.

4.2 How AgentShield Enforces Security

AgentShield operates at two distinct pre-execution checkpoints:

- 1. **Semantic AST Inspection**: Intercepts tool calls and shell strings *before* process spawning. Parses command syntax trees to detect dangerous tokens and blacklisted operations.
- 2. **Linux Landlock LSM Kernel Isolation**: Unprivileged Landlock security modules lock down filesystem access, strictly confining agent subprocesses to safe scratch directories (e.g. `/tmp/scratch` and the current workspace).



Figure 10.1: Microsoft 365 Copilot & Enterprise Power Platform Integration Topology

Module 5: PILOT Just-In-Time Multi-Tier Memory

5.1 The 3-Tier Cognitive Memory Hierarchy

Tagisan organises memory into three purpose-built storage tiers:

| Memory Tier | Backing Store | Lifetime | Access Latency | Primary Role |

| :--- | :--- | :--- | :--- | :--- |

| **Tier 1: Working Memory** | Tokio In-Memory RAM | In-flight mission turn | < 1 millisecond | Active conversation scratchpad, scratch variables |

| **Tier 2: Episodic Memory** | SQLite / DuckDB | Persistent across sessions | < 5 milliseconds | Complete session audit logs, past tool outputs, rollbacks |

| **Tier 3: Semantic Memory** | Qdrant / SQLite-vec | Indefinite persistent | < 15 milliseconds | Vector embeddings, code graph indices, domain knowledge |

5.2 Hybrid Retrieval with Reciprocal Rank Fusion (RRF)

To overcome vector-only semantic drift and keyword-only lexical blindness, Tagisan combines dense neural embeddings with sparse BM25 token indices using **Reciprocal Rank Fusion (RRF)**:

$$RRF(d) = \sum_{m \in M} \frac{1}{k + r_m(d)}$$

Where $k = 60$ and $r_m(d)$ is the rank of document d within retriever m . This mathematical fusion ensures that exact symbol matches (e.g., function names, variable identifiers) and high-level conceptual descriptions receive optimal relevance scoring.

Module 6: Swarm MoA & Hegelian Dialectical Debate Engine

6.1 Multi-Agent Swarm Mixture-of-Agents (MoA)

Single-agent architectures suffer from confirmation bias and catastrophic drift on long reasoning chains. Tagisan orchestrates a 4-agent dialectical swarm:

1. **Thesis Proposer**: Formulates the initial technical proposal or implementation plan.
 2. **Antithesis Challenger**: Rigorously audits the proposal for edge cases, security vulnerabilities, performance regressions, and hiring/maintenance costs.
 3. **Synthesis Reconciler**: Merges valid criticisms into an improved, battle-tested compromise architecture.
 4. **Judge / Arbiter**: Performs mathematical and formal verification, evaluates cross-examinations, and ratifies the final binding decision.
-

Module 7: The 500-Skill Sovereign Catalog (RFC-004)

7.1 The RFC-004 Dynamic Skill Standard

Every skill in Tagisan is a self-contained, versioned capability defined under `.ecc/skills//SKILL.md` with structured YAML frontmatter:

```
---
name: postgresql-zero-downtime-migrator
description: Plans and executes zero-downtime PostgreSQL schema migrations with lock timeout
guards.
version: 1.0.0
tags: [database, postgresql, dba, performance]
runtime: rust
allow_network: false
entrypoint: src/migrator.rs
---

# Sovereign Agent Instructions
1. Always check pg_stat_activity before executing DDL.
2. Set lock_timeout = '2s' before acquiring ACCESS EXCLUSIVE locks.
3. Use CREATE INDEX CONCURRENTLY for all index additions.
```

7.2 Managing and Chaining Skills

Skills can be dynamically chained together in complex multi-step missions:

```
# Invoke chained skills in a single pipeline
tgs run --skill "k8s-pod-diagnostics,postgresql-zero-downtime-migrator" \
  "Diagnose database connection timeouts and optimize connection pooling"
```

Module 8: Model Context Protocol (MCP 500)

Enterprise Hub

8.1 The Model Context Protocol Standard

The Model Context Protocol (MCP) standardizes how AI agents discover and execute external tools over JSON-RPC 2.0. Tagisan implements both **stdio** (sub-process) and **SSE** (Server-Sent Events) transports, equipped with a curated catalog of **500 enterprise plugins** sourced exclusively from verified registries (Smithery.ai, NPM, PyPI, Glama.ai, Composio, Cloudflare).

8.2 The 10 Strategic MCP Domains

- Search, Web Scraping & Deep Research** (50 plugins): Brave Search, Firecrawl, Tavily, Puppeteer.
 - Relational Databases, OLAP & Event Streaming** (60 plugins): PostgreSQL, MySQL, ClickHouse, Kafka, Snowflake.
 - Vector Databases & Semantic Memory** (40 plugins): Qdrant, Pinecone, Milvus, Chroma, Weaviate.
 - Cloud Infrastructure, Serverless & Edge** (55 plugins): AWS, GCP, Azure, Cloudflare Workers, Terraform.
 - DevOps, Containers & Kubernetes Orchestration** (50 plugins): Kubernetes, Docker, Helm, ArgoCD, Ansible.
 - Observability, APM & SRE Reliability** (45 plugins): Prometheus, Grafana, Datadog, Jaeger, OpenTelemetry.
 - Cybersecurity, EDR & Threat Defense** (50 plugins): Trivy, Slither, VirusTotal, Shodan, Wireshark.
 - Developer Productivity & Tracking** (50 plugins): GitHub, GitLab, Jira, Linear, Slack, Notion.
 - Enterprise SaaS, CRM & FinTech** (50 plugins): Salesforce, Stripe, HubSpot, QuickBooks, SAP.
 - AI Models, Multimodal & GPU Compute** (50 plugins): Ollama, HuggingFace, Replicate, vLLM, DeepSeek.
-

Module 9: Domain Copilots & Digital Twin Engineering (VELLA)

9.1 VELLA: The Cyber-Physical Digital Twin Engine

VELLA bridges LLM cognitive reasoning with deterministic real-time cyber-physical systems:



Figure 10.1: Microsoft 365 Copilot & Enterprise Power Platform Integration Topology

9.2 Live VELLA CLI Commands

```
# Industrial SCADA telemetry inspection
tgs vella scada --endpoint "tcp://192.168.1.100:502" --analog 85.4 --alarm "trip_cooling"

# Quantitative Forex margin and pip calculation
tgs vella forex --pair "EUR/USD" --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0

# Aerospace satellite SGP4 orbit propagation
tgs vella aerospace --minutes 90.0 --tle "1 25544U..."

# Genomic sequence alignment
tgs vella bio --target "ACTGATCG" --template "ACTGATCG" --ref-genome "GRCh38"
```

Module 10: Enterprise Hardening, Air-Gapping & Observability

10.1 Zero-Trust Air-Gapped Deployment

In classified or banking environments with zero outbound internet access:

1. Compile `tagisan` release binary: `cargo build --release`.
2. Set environment privacy lock: `export TAGISAN\_LOCAL\_ONLY=1` and `export TAGISAN\_OFFLINE=1`.
3. Deploy local Ollama with quantized GGUF weights (`smollm2`, `qwen2.5-coder`).
4. AgentShield enforces complete offline sandboxing, guaranteeing zero network bytes leave the server rack.

10.2 Observability & Cost Accounting

Every Tagisan execution outputs structured OpenTelemetry JSON records tracking:

- `correlation\_id` and execution span hierarchy.
 - Exact wall-clock latency (ms).
 - Prompt tokens, completion tokens, and cached prompt tokens.
 - Exact monetary expenditure in USD (\$0.0000 on Google OAuth and local Ollama).
-

Module 11: The Definitive 950 Modern Environment Scenarios

This catalog documents **950** concrete, production-proven scenarios organized across 19 strategic enterprise engineering domains (50 scenarios per domain). Every scenario details the exact operational objective, TGS capabilities leveraged, the command syntax, the automated multi-step execution flow, and the verifiable sovereign outcome.

Domain 1–50: Cloud Architecture, Infrastructure as Code & Multi-Cloud

Scenario 1: Terraform State Drift Auto-Reconciliation: Enterprise Baseline

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Production baseline operational state verified with zero drift.

Scenario 2: Multi-Cloud Failover Route53 to Cloudflare DNS: Enterprise Baseline

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.

2. Flips Route53 CNAME to GCP GKE ingress.

3. Verifies read replica promotion.

4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Production baseline operational state verified with zero drift.

Scenario 3: AWS IAM Least-Privilege Role Pruner: Enterprise Baseline

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.

2. Discovers unused s3:* and sqs:* wildcards.

3. Emits scoped least-privilege IAM policy.

4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Production baseline operational state verified with zero drift.

Scenario 4: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Enterprise Baseline

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Production baseline operational state verified with zero drift.

Scenario 5: Azure Bicep to Terraform HCL Transpiler: Enterprise Baseline

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerem.
3. Validates syntactically correct HCL with terraform validate.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Production baseline operational state verified with zero drift.

Scenario 6: Terraform State Drift Auto-Reconciliation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 7: Multi-Cloud Failover Route53 to Cloudflare DNS: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 8: AWS IAM Least-Privilege Role Pruner: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 9: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 10: Azure Bicep to Terraform HCL Transpiler: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 11: Terraform State Drift Auto-Reconciliation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 12: Multi-Cloud Failover Route53 to Cloudflare DNS: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 13: AWS IAM Least-Privilege Role Pruner: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 14: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 15: Azure Bicep to Terraform HCL Transpiler: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 16: Terraform State Drift Auto-Reconciliation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 17: Multi-Cloud Failover Route53 to Cloudflare DNS: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 18: AWS IAM Least-Privilege Role Pruner: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 19: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 20: Azure Bicep to Terraform HCL Transpiler: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 21: Terraform State Drift Auto-Reconciliation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 22: Multi-Cloud Failover Route53 to Cloudflare DNS: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 23: AWS IAM Least-Privilege Role Pruner: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 24: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 25: Azure Bicep to Terraform HCL Transpiler: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 26: Terraform State Drift Auto-Reconciliation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 27: Multi-Cloud Failover Route53 to Cloudflare DNS: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 28: AWS IAM Least-Privilege Role Pruner: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 29: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 30: Azure Bicep to Terraform HCL Transpiler: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 31: Terraform State Drift Auto-Reconciliation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 32: Multi-Cloud Failover Route53 to Cloudflare DNS: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 33: AWS IAM Least-Privilege Role Pruner: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 34: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 35: Azure Bicep to Terraform HCL Transpiler: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 36: Terraform State Drift Auto-Reconciliation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 37: Multi-Cloud Failover Route53 to Cloudflare DNS: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 38: AWS IAM Least-Privilege Role Pruner: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 39: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 40: Azure Bicep to Terraform HCL Transpiler: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 41: Terraform State Drift Auto-Reconciliation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
2. Flags manually edited security groups.
3. Applies verified least-privilege rules.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 42: Multi-Cloud Failover Route53 to Cloudflare DNS: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
2. Flips Route53 CNAME to GCP GKE ingress.
3. Verifies read replica promotion.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 43: AWS IAM Least-Privilege Role Pruner: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.
2. Discovers unused s3:* and sqs:* wildcards.
3. Emits scoped least-privilege IAM policy.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 44: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB).
3. Snapshots volumes and terminates zombie clusters.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 45: Azure Bicep to Terraform HCL Transpiler: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
2. Maps Azure resource types to terraform-provider-azurerm.
3. Validates syntactically correct HCL with terraform validate.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 46: Terraform State Drift Auto-Reconciliation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Terraform MCP, Swarm MoA, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Detect and reconcile state drift in main.tf across AWS us-east-1 and us-west-2"
```

```
```
```

- **Execution Flow**:

1. Runs terraform plan.
 2. Flags manually edited security groups.
 3. Applies verified least-privilege rules.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Infrastructure state reconciled with zero downtime. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 47: Multi-Cloud Failover Route53 to Cloudflare DNS: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `DNS Routing MCP, Gemini 3 Flash, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Simulate AWS edge outage and execute automated DNS traffic cutover to GCP backup"
```

```
```
```

- **Execution Flow**:

1. Health check detects 100% packet loss in AWS.
 2. Flips Route53 CNAME to GCP GKE ingress.
 3. Verifies read replica promotion.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Complete traffic rerouted in 45 seconds. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 48: AWS IAM Least-Privilege Role Pruner: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AWS IAM MCP, AgentShield, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Analyze CloudTrail 90-day logs for role api-worker and prune unused permissions"
```

```
```
```

- **Execution Flow**:

1. Ingests CloudTrail AssumeRole events.

2. Discovers unused s3:* and sqs:* wildcards.

3. Emits scoped least-privilege IAM policy.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Attack surface reduced by 85% with zero broken workloads. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 49: Cloud Cost Anomaly Hunter: Idle EBS & EKS: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Cloud Cost MCP, SQLite Memory, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters"
```

```
```
```

- **Execution Flow**:

1. Scans EC2/EKS metrics for zero CPU utilization.

2. Discovers 14 unattached EBS volumes (8.4 TB).

3. Snapshots volumes and terminates zombie clusters.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: \$33,600 annual cloud savings realized immediately. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 50: Azure Bicep to Terraform HCL Transpiler: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AST Transpiler Engine, Gemini 2.5 Pro, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Transpile azuredeploy.bicep into modular Terraform HCL with AzAPI provider"
```

```
```
```

- **Execution Flow**:

1. Parses Bicep AST resources and parameters.
 2. Maps Azure resource types to terraform-provider-azurerm.
 3. Validates syntactically correct HCL with terraform validate.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Complete infrastructure codebase migrated in 12 seconds. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 51–100: Kubernetes, Containerization & Microservices Orchestration

Scenario 51: Kubernetes CrashLoopBackOff Auto-Diagnosis: Enterprise Baseline

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
 2. Pinpoints OOMKilled condition.
 3. Rolls back Helm release to previous stable revision.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Pod restored to running status in 30s. Production baseline operational state verified with zero drift.

Scenario 52: Cilium eBPF Network Policy Enforcement: Enterprise Baseline

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Production baseline operational state verified with zero drift.

Scenario 53: ArgoCD GitOps Sync Failure Healer: Enterprise Baseline

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Production baseline operational state verified with zero drift.

Scenario 54: Karpenter Node Autoscaler Consolidation Tuner: Enterprise Baseline

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Production baseline operational state verified with zero drift.

Scenario 55: Istio mTLS Certificate Expiry Auto-Rotator: Enterprise Baseline

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Production baseline operational state verified with zero drift.

Scenario 56: Kubernetes CrashLoopBackOff Auto-Diagnosis: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Pod restored to running status in 30s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 57: Cilium eBPF Network Policy Enforcement: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 58: ArgoCD GitOps Sync Failure Healer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 59: Karpenter Node Autoscaler Consolidation Tuner: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 60: Istio mTLS Certificate Expiry Auto-Rotator: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
 2. Flags 3 sidecars with stale SDS tokens.
 3. Restarts Envoy proxies with zero dropped active connections.
 4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
 5. Executes automated health check cutover validation.
- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 61: Kubernetes CrashLoopBackOff Auto-Diagnosis: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
 2. Pinpoints OOMKilled condition.
 3. Rolls back Helm release to previous stable revision.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Pod restored to running status in 30s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 62: Cilium eBPF Network Policy Enforcement: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
 2. Formulates CiliumNetworkPolicy CRD.
 3. Verifies unauthorized egress packets dropped at kernel layer.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 63: ArgoCD GitOps Sync Failure Healer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
 2. Identifies deprecated autoscaling/v2beta1 API.
 3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 64: Karpenter Node Autoscaler Consolidation Tuner: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 65: Istio mTLS Certificate Expiry Auto-Rotator: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 66: Kubernetes CrashLoopBackOff Auto-Diagnosis: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Pod restored to running status in 30s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 67: Cilium eBPF Network Policy Enforcement: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 68: ArgoCD GitOps Sync Failure Healer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 69: Karpenter Node Autoscaler Consolidation Tuner: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 70: Istio mTLS Certificate Expiry Auto-Rotator: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 71: Kubernetes CrashLoopBackOff Auto-Diagnosis: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Pod restored to running status in 30s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 72: Cilium eBPF Network Policy Enforcement: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
 2. Formulates CiliumNetworkPolicy CRD.
 3. Verifies unauthorized egress packets dropped at kernel layer.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Zero-trust container network segmentation enforced. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 73: ArgoCD GitOps Sync Failure Healer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
 2. Identifies deprecated autoscaling/v2beta1 API.
 3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 74: Karpenter Node Autoscaler Consolidation Tuner: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
 2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
 3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 75: Istio mTLS Certificate Expiry Auto-Rotator: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
 2. Flags 3 sidecars with stale SDS tokens.
 3. Restarts Envoy proxies with zero dropped active connections.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 76: Kubernetes CrashLoopBackOff Auto-Diagnosis: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Pod restored to running status in 30s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 77: Cilium eBPF Network Policy Enforcement: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 78: ArgoCD GitOps Sync Failure Healer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 79: Karpenter Node Autoscaler Consolidation Tuner: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 80: Istio mTLS Certificate Expiry Auto-Rotator: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 81: Kubernetes CrashLoopBackOff Auto-Diagnosis: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Pod restored to running status in 30s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 82: Cilium eBPF Network Policy Enforcement: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 83: ArgoCD GitOps Sync Failure Healer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 84: Karpenter Node Autoscaler Consolidation Tuner: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 85: Istio mTLS Certificate Expiry Auto-Rotator: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 86: Kubernetes CrashLoopBackOff Auto-Diagnosis: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Pod restored to running status in 30s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 87: Cilium eBPF Network Policy Enforcement: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 88: ArgoCD GitOps Sync Failure Healer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
 2. Identifies deprecated autoscaling/v2beta1 API.
 3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 89: Karpenter Node Autoscaler Consolidation Tuner: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
 2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
 3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 90: Istio mTLS Certificate Expiry Auto-Rotator: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 91: Kubernetes CrashLoopBackOff Auto-Diagnosis: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
2. Pinpoints OOMKilled condition.
3. Rolls back Helm release to previous stable revision.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Pod restored to running status in 30s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 92: Cilium eBPF Network Policy Enforcement: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
2. Formulates CiliumNetworkPolicy CRD.
3. Verifies unauthorized egress packets dropped at kernel layer.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 93: ArgoCD GitOps Sync Failure Healer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
2. Identifies deprecated autoscaling/v2beta1 API.
3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 94: Karpenter Node Autoscaler Consolidation Tuner: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 95: Istio mTLS Certificate Expiry Auto-Rotator: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.
2. Flags 3 sidecars with stale SDS tokens.
3. Restarts Envoy proxies with zero dropped active connections.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 96: Kubernetes CrashLoopBackOff Auto-Diagnosis: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Kubernetes MCP, AgentShield, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Diagnose CrashLoopBackOff in pod auth-svc in prod and restore service"
```

```
```
```

- **Execution Flow**:

1. Queries kube-apiserver events.
 2. Pinpoints OOMKilled condition.
 3. Rolls back Helm release to previous stable revision.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Pod restored to running status in 30s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 97: Cilium eBPF Network Policy Enforcement: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Cilium MCP, eBPF Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Deploy Cilium NetworkPolicy restricting pod database access to backend namespace only"
```

```
```
```

- **Execution Flow**:

1. Analyzes L7 HTTP and L3/L4 traffic flows.
 2. Formulates CiliumNetworkPolicy CRD.
 3. Verifies unauthorized egress packets dropped at kernel layer.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Zero-trust container network segmentation enforced. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 98: ArgoCD GitOps Sync Failure Healer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `GitOps MCP, Dialectical Debate, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Resolve OutOfSync degraded state on ArgoCD app payment-service"
```

```
```
```

- **Execution Flow**:

1. Inspects live cluster diff vs git manifest.
 2. Identifies deprecated autoscaling/v2beta1 API.
 3. Auto-upgrades manifest to autoscaling/v2 and triggers sync.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: ArgoCD app restored to Synced and Healthy in 40s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 99: Karpenter Node Autoscaler Consolidation Tuner: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AWS Karpenter MCP, SRE Engine, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Tune Karpenter consolidation policy to pack idle micro-instances onto Graviton spot nodes"
```

```
```
```

- **Execution Flow**:

1. Analyzes pod resource requests vs node allocations.
 2. Configures consolidationPolicy: WhenEmptyOrUnderutilized.
 3. Evicts non-critical pods smoothly using PodDisruptionBudgets.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Cluster EC2 instance count reduced by 48%. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 100: Istio mTLS Certificate Expiry Auto-Rotator: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Istio MCP, OpenSSL Toolchain, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Audit all workload certificates expiring in 72h and trigger Citadel secret rotation"
```

```
```
```

- **Execution Flow**:

1. Dumps Envoy TLS certificates across 120 pods.

2. Flags 3 sidecars with stale SDS tokens.

3. Restarts Envoy proxies with zero dropped active connections.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: 100% mTLS certificate renewal completed seamlessly. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 101–150: Site Reliability Engineering, Observability & Chaos

Scenario 101: Prometheus Alert Fatigue Clusterer: Enterprise Baseline

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.

2. Groups 442 downstream HTTP 504 alerts using vector similarity.

3. Pinpoints primary Redis connection starvation.

4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Production baseline operational state verified with zero drift.

Scenario 102: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Enterprise Baseline

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Production baseline operational state verified with zero drift.

Scenario 103: Chaos Mesh Network Latency Injection Test: Enterprise Baseline

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Production baseline operational state verified with zero drift.

Scenario 104: Automated Incident Post-Mortem Synthesizer: Enterprise Baseline

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Production baseline operational state verified with zero drift.

Scenario 105: eBPF On-CPU Kernel Profiler with FlameGraphs: Enterprise Baseline

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Production baseline operational state verified with zero drift.

Scenario 106: Prometheus Alert Fatigue Clusterer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
 2. Groups 442 downstream HTTP 504 alerts using vector similarity.
 3. Pinpoints primary Redis connection starvation.
 4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
 5. Executes automated health check cutover validation.
- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 107: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
 2. Detects unindexed SQL query inside coupon validation service.
 3. Synthesizes migration creating index on coupon_code.
 4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
 5. Executes automated health check cutover validation.
- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 108: Chaos Mesh Network Latency Injection Test: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 109: Automated Incident Post-Mortem Synthesizer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 110: eBPF On-CPU Kernel Profiler with FlameGraphs: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 111: Prometheus Alert Fatigue Clusterer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 112: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 113: Chaos Mesh Network Latency Injection Test: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 114: Automated Incident Post-Mortem Synthesizer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
 2. Structures timeline from Detection to Mitigation.
 3. Generates executive summary and 5 preventative action items.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 115: eBPF On-CPU Kernel Profiler with FlameGraphs: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
 2. Collects 1,000,000 stack traces.
 3. Generates interactive SVG FlameGraph highlighting regex backtracking.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 116: Prometheus Alert Fatigue Clusterer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 117: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 118: Chaos Mesh Network Latency Injection Test: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 119: Automated Incident Post-Mortem Synthesizer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 120: eBPF On-CPU Kernel Profiler with FlameGraphs: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 121: Prometheus Alert Fatigue Clusterer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 122: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 123: Chaos Mesh Network Latency Injection Test: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 124: Automated Incident Post-Mortem Synthesizer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 125: eBPF On-CPU Kernel Profiler with FlameGraphs: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 126: Prometheus Alert Fatigue Clusterer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 127: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 128: Chaos Mesh Network Latency Injection Test: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 129: Automated Incident Post-Mortem Synthesizer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 130: eBPF On-CPU Kernel Profiler with FlameGraphs: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 131: Prometheus Alert Fatigue Clusterer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 132: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 133: Chaos Mesh Network Latency Injection Test: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 134: Automated Incident Post-Mortem Synthesizer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 135: eBPF On-CPU Kernel Profiler with FlameGraphs: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 136: Prometheus Alert Fatigue Clusterer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
 2. Groups 442 downstream HTTP 504 alerts using vector similarity.
 3. Pinpoints primary Redis connection starvation.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 137: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
 2. Detects unindexed SQL query inside coupon validation service.
 3. Synthesizes migration creating index on coupon_code.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 138: Chaos Mesh Network Latency Injection Test: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 139: Automated Incident Post-Mortem Synthesizer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 140: eBPF On-CPU Kernel Profiler with FlameGraphs: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 141: Prometheus Alert Fatigue Clusterer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
2. Groups 442 downstream HTTP 504 alerts using vector similarity.
3. Pinpoints primary Redis connection starvation.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 142: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
2. Detects unindexed SQL query inside coupon validation service.
3. Synthesizes migration creating index on coupon_code.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 143: Chaos Mesh Network Latency Injection Test: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
2. Monitors Envoy circuit breaker state.
3. Confirms circuit breaker trips and falls back to cached response.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 144: Automated Incident Post-Mortem Synthesizer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
2. Structures timeline from Detection to Mitigation.
3. Generates executive summary and 5 preventative action items.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 145: eBPF On-CPU Kernel Profiler with FlameGraphs: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
2. Collects 1,000,000 stack traces.
3. Generates interactive SVG FlameGraph highlighting regex backtracking.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 146: Prometheus Alert Fatigue Clusterer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Prometheus MCP, PILOT Memory, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Cluster 450 firing Alertmanager alerts from incident #8821 to identify primary root cause"
```

```
```
```

- **Execution Flow**:

1. Ingests raw Alertmanager JSON payload.
 2. Groups 442 downstream HTTP 504 alerts using vector similarity.
 3. Pinpoints primary Redis connection starvation.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: On-call alerted to 1 actionable root cause instead of 450. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 147: OpenTelemetry Distributed Trace Bottleneck Pinpointer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Jaeger/Otel MCP, Gemini 3.1 Pro, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Analyze 1,000 p99 traces on /checkout endpoint and isolate latency spike sources"
```

```
```
```

- **Execution Flow**:

1. Traverses DAG call graph across 14 microservices.
 2. Detects unindexed SQL query inside coupon validation service.
 3. Synthesizes migration creating index on coupon_code.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Checkout latency reduced from 3.4s to 110ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 148: Chaos Mesh Network Latency Injection Test: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Chaos Mesh MCP, BEAM Actor Supervisor, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Inject 250ms packet latency on payment-gateway and verify circuit breaker trip"
```

```
```
```

- **Execution Flow**:

1. Deploys Chaos Mesh NetworkChaos CRD.
 2. Monitors Envoy circuit breaker state.
 3. Confirms circuit breaker trips and falls back to cached response.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Zero cascading microservice failure during network degradation. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 149: Automated Incident Post-Mortem Synthesizer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `PILOT Episodic Memory, Gemini Pro, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty logs"
```

```
```
```

- **Execution Flow**:

1. Ingests chat logs, alert timestamps, and git commits.
 2. Structures timeline from Detection to Mitigation.
 3. Generates executive summary and 5 preventative action items.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Executive post-mortem document generated in markdown. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 150: eBPF On-CPU Kernel Profiler with FlameGraphs: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `eBPF Profiler, Rust Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Profile CPU spikes on worker node #4 and output SVG FlameGraph identifying hot functions"
```

```
```
```

- **Execution Flow**:

1. Attaches eBPF sampling probe to kernel sched_switch.
 2. Collects 1,000,000 stack traces.
 3. Generates interactive SVG FlameGraph highlighting regex backtracking.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Regex hot-path refactored, dropping CPU usage by 65%. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 151–200: Cybersecurity, Zero-Trust Architecture & Threat Hunting

Scenario 151: Linux Landlock LSM Kernel Jail Enforcer: Enterprise Baseline

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
 2. Strips raw socket network capabilities.
 3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Production baseline operational state verified with zero drift.

Scenario 152: Active Directory Pass-the-Hash Hunter: Enterprise Baseline

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Production baseline operational state verified with zero drift.

Scenario 153: Ransomware Mass Encryption Detector & Storage Freezer: Enterprise Baseline

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Production baseline operational state verified with zero drift.

Scenario 154: Obfuscated Malicious Bash Deobfuscator: Enterprise Baseline

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Production baseline operational state verified with zero drift.

Scenario 155: Cloudflare WAF JA3 Fingerprint Rule Generator: Enterprise Baseline

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
2. Computes common TLS ClientHello JA3 hash.
3. Deploys Cloudflare custom rule blocking fingerprint.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Production baseline operational state verified with zero drift.

Scenario 156: Linux Landlock LSM Kernel Jail Enforcer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
 2. Strips raw socket network capabilities.
 3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
 4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
 5. Executes automated health check cutover validation.
- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 157: Active Directory Pass-the-Hash Hunter: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
 2. Flags NTLM authentication anomaly from unmanaged workstation.
 3. Correlates to PsExec remote service creation.
 4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
 5. Executes automated health check cutover validation.
- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 158: Ransomware Mass Encryption Detector & Storage Freezer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 159: Obfuscated Malicious Bash Deobfuscator: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 160: Cloudflare WAF JA3 Fingerprint Rule Generator: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
2. Computes common TLS ClientHello JA3 hash.
3. Deploys Cloudflare custom rule blocking fingerprint.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 161: Linux Landlock LSM Kernel Jail Enforcer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
2. Strips raw socket network capabilities.
3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 162: Active Directory Pass-the-Hash Hunter: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 163: Ransomware Mass Encryption Detector & Storage Freezer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 164: Obfuscated Malicious Bash Deobfuscator: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.

2. Discovers persistence cron job downloading cryptominer.

3. Extracts C2 IP addresses and file hashes.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 165: Cloudflare WAF JA3 Fingerprint Rule Generator: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.

2. Computes common TLS ClientHello JA3 hash.

3. Deploys Cloudflare custom rule blocking fingerprint.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 166: Linux Landlock LSM Kernel Jail Enforcer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
2. Strips raw socket network capabilities.
3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 167: Active Directory Pass-the-Hash Hunter: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 168: Ransomware Mass Encryption Detector & Storage Freezer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 169: Obfuscated Malicious Bash Deobfuscator: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 170: Cloudflare WAF JA3 Fingerprint Rule Generator: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
2. Computes common TLS ClientHello JA3 hash.
3. Deploys Cloudflare custom rule blocking fingerprint.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 171: Linux Landlock LSM Kernel Jail Enforcer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
2. Strips raw socket network capabilities.
3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 172: Active Directory Pass-the-Hash Hunter: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 173: Ransomware Mass Encryption Detector & Storage Freezer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 174: Obfuscated Malicious Bash Deobfuscator: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
 2. Discovers persistence cron job downloading cryptominer.
 3. Extracts C2 IP addresses and file hashes.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Full IOC threat intelligence report generated. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 175: Cloudflare WAF JA3 Fingerprint Rule Generator: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
 2. Computes common TLS ClientHello JA3 hash.
 3. Deploys Cloudflare custom rule blocking fingerprint.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 176: Linux Landlock LSM Kernel Jail Enforcer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
2. Strips raw socket network capabilities.
3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 177: Active Directory Pass-the-Hash Hunter: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 178: Ransomware Mass Encryption Detector & Storage Freezer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 179: Obfuscated Malicious Bash Deobfuscator: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 180: Cloudflare WAF JA3 Fingerprint Rule Generator: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
2. Computes common TLS ClientHello JA3 hash.
3. Deploys Cloudflare custom rule blocking fingerprint.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 181: Linux Landlock LSM Kernel Jail Enforcer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
2. Strips raw socket network capabilities.
3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 182: Active Directory Pass-the-Hash Hunter: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 183: Ransomware Mass Encryption Detector & Storage Freezer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 184: Obfuscated Malicious Bash Deobfuscator: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
 2. Discovers persistence cron job downloading cryptominer.
 3. Extracts C2 IP addresses and file hashes.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: Full IOC threat intelligence report generated. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 185: Cloudflare WAF JA3 Fingerprint Rule Generator: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
 2. Computes common TLS ClientHello JA3 hash.
 3. Deploys Cloudflare custom rule blocking fingerprint.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 186: Linux Landlock LSM Kernel Jail Enforcer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
 2. Strips raw socket network capabilities.
 3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 187: Active Directory Pass-the-Hash Hunter: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
 2. Flags NTLM authentication anomaly from unmanaged workstation.
 3. Correlates to PsExec remote service creation.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 188: Ransomware Mass Encryption Detector & Storage Freezer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 189: Obfuscated Malicious Bash Deobfuscator: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 190: Cloudflare WAF JA3 Fingerprint Rule Generator: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
 2. Computes common TLS ClientHello JA3 hash.
 3. Deploys Cloudflare custom rule blocking fingerprint.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 191: Linux Landlock LSM Kernel Jail Enforcer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
 2. Strips raw socket network capabilities.
 3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 192: Active Directory Pass-the-Hash Hunter: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
2. Flags NTLM authentication anomaly from unmanaged workstation.
3. Correlates to PsExec remote service creation.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 193: Ransomware Mass Encryption Detector & Storage Freezer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.
2. Shannon entropy calculation confirms encryption.
3. Revokes NFS client credentials and freezes Ceph snapshot.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 194: Obfuscated Malicious Bash Deobfuscator: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.
2. Discovers persistence cron job downloading cryptominer.
3. Extracts C2 IP addresses and file hashes.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 195: Cloudflare WAF JA3 Fingerprint Rule Generator: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
2. Computes common TLS ClientHello JA3 hash.
3. Deploys Cloudflare custom rule blocking fingerprint.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 196: Linux Landlock LSM Kernel Jail Enforcer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Landlock LSM Kernel Sandboxing, AgentShield, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Execute untrusted scraper binary and restrict access strictly to /tmp/scratch"
```

```
```
```

- **Execution Flow**:

1. Configures Landlock rules restricting /etc and /home.
 2. Strips raw socket network capabilities.
 3. Intercepts attempt to read /etc/passwd with immediate SIGKILL.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Zero-trust kernel sandbox contained threat completely. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 197: Active Directory Pass-the-Hash Hunter: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Windows Event Parser, Vector Memory, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Analyze EventID 4624 Type 3 logs to detect lateral movement across domain controllers"
```

```
```
```

- **Execution Flow**:

1. Ingests 500,000 authentication logs.
 2. Flags NTLM authentication anomaly from unmanaged workstation.
 3. Correlates to PsExec remote service creation.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Compromised machine isolated from Active Directory in 3m. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 198: Ransomware Mass Encryption Detector & Storage Freezer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AgentShield I/O Watcher, Ceph MCP, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Monitor /mnt/storage for rapid file renaming and Shannon entropy spikes"
```

```
```
```

- **Execution Flow**:

1. Measures 1,200 renames/sec to .locked.

2. Shannon entropy calculation confirms encryption.

3. Revokes NFS client credentials and freezes Ceph snapshot.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: 99.4% of corporate storage preserved via instant snapshot. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 199: Obfuscated Malicious Bash Deobfuscator: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AgentShield AST Deobfuscator, Local Ollama, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Deobfuscate nested base64 pipe-to-bash script intercepted on honeypot"
```

```
```
```

- **Execution Flow**:

1. Unpacks nested base64, gzip, and rot13 without shell execution.

2. Discovers persistence cron job downloading cryptominer.

3. Extracts C2 IP addresses and file hashes.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Full IOC threat intelligence report generated. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 200: Cloudflare WAF JA3 Fingerprint Rule Generator: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Cloudflare WAF MCP, Regex Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgts debate --rounds 3 "Analyze access.log for bot scraper patterns and deploy JA3 fingerprint rule"
```

```
```
```

- **Execution Flow**:

1. Identifies 400 rotating residential proxies.
 2. Computes common TLS ClientHello JA3 hash.
 3. Deploys Cloudflare custom rule blocking fingerprint.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Bot traffic reduced from 92% to 0.01%. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 201–250: DevSecOps, CI/CD Pipeline & Supply Chain Security

Scenario 201: Git Secret Scrubbing with BFG Repo-Cleaner: Enterprise Baseline

- **TGS Capabilities**: `Git History Engine, AgentShield, Production Baseline`

- **Command**:

```
```bash
```

```
tgts run "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
 2. Discovers exposed secret committed in old config.
 3. Invokes BFG cleaner, forces push, and rotates IAM key.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Production baseline operational state verified with zero drift.

Scenario 202: Container SBOM Generator & Trivy CVE Matcher: Enterprise Baseline

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Production baseline operational state verified with zero drift.

Scenario 203: Cosign Container Image Signature & Rekor Attestation: Enterprise Baseline

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Production baseline operational state verified with zero drift.

Scenario 204: Semgrep Static Analysis Rule Synthesizer: Enterprise Baseline

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Production baseline operational state verified with zero drift.

Scenario 205: GitHub Actions Workflow Hardening against PwnPR: Enterprise Baseline

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Production baseline operational state verified with zero drift.

Scenario 206: Git Secret Scrubbing with BFG Repo-Cleaner: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Git History Engine, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 207: Container SBOM Generator & Trivy CVE Matcher: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 208: Cosign Container Image Signature & Rekor Attestation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 209: Semgrep Static Analysis Rule Synthesizer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Semgrep MCP, AST Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 210: GitHub Actions Workflow Hardening against PwnPR: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 211: Git Secret Scrubbing with BFG Repo-Cleaner: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Git History Engine, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 212: Container SBOM Generator & Trivy CVE Matcher: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 213: Cosign Container Image Signature & Rekor Attestation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 214: Semgrep Static Analysis Rule Synthesizer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Semgrep MCP, AST Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 215: GitHub Actions Workflow Hardening against PwnPR: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 216: Git Secret Scrubbing with BFG Repo-Cleaner: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Git History Engine, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 217: Container SBOM Generator & Trivy CVE Matcher: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 218: Cosign Container Image Signature & Rekor Attestation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 219: Semgrep Static Analysis Rule Synthesizer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 220: GitHub Actions Workflow Hardening against PwnPR: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 221: Git Secret Scrubbing with BFG Repo-Cleaner: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Git History Engine, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 222: Container SBOM Generator & Trivy CVE Matcher: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 223: Cosign Container Image Signature & Rekor Attestation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 224: Semgrep Static Analysis Rule Synthesizer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Semgrep MCP, AST Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
 2. Discovers 4 raw format strings in database DAO.
 3. Rewrites queries using SQLAlchemy parameter bindings.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 225: GitHub Actions Workflow Hardening against PwnPR: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
 2. Flags pull_request_target with checkout of untrusted PR head.
 3. Rewrites workflow to pull_request with read-only token permissions.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 226: Git Secret Scrubbing with BFG Repo-Cleaner: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Git History Engine, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 227: Container SBOM Generator & Trivy CVE Matcher: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 228: Cosign Container Image Signature & Rekor Attestation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 229: Semgrep Static Analysis Rule Synthesizer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 230: GitHub Actions Workflow Hardening against PwnPR: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 231: Git Secret Scrubbing with BFG Repo-Cleaner: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Git History Engine, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 232: Container SBOM Generator & Trivy CVE Matcher: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.

2. Cross-references NVD database.

3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.

4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.

5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 233: Cosign Container Image Signature & Rekor Attestation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.

2. Signs container image digest.

3. Publishes attestation to Rekor transparency log.

4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.

5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 234: Semgrep Static Analysis Rule Synthesizer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 235: GitHub Actions Workflow Hardening against PwnPR: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 236: Git Secret Scrubbing with BFG Repo-Cleaner: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Git History Engine, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 237: Container SBOM Generator & Trivy CVE Matcher: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
2. Cross-references NVD database.
3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 238: Cosign Container Image Signature & Rekor Attestation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
2. Signs container image digest.
3. Publishes attestation to Rekor transparency log.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 239: Semgrep Static Analysis Rule Synthesizer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Semgrep MCP, AST Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 240: GitHub Actions Workflow Hardening against PwnPR: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 241: Git Secret Scrubbing with BFG Repo-Cleaner: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Git History Engine, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
2. Discovers exposed secret committed in old config.
3. Invokes BFG cleaner, forces push, and rotates IAM key.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 242: Container SBOM Generator & Trivy CVE Matcher: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
 2. Cross-references NVD database.
 3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 243: Cosign Container Image Signature & Rekor Attestation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.
 2. Signs container image digest.
 3. Publishes attestation to Rekor transparency log.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 244: Semgrep Static Analysis Rule Synthesizer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.
2. Discovers 4 raw format strings in database DAO.
3. Rewrites queries using SQLAlchemy parameter bindings.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 245: GitHub Actions Workflow Hardening against PwnPR: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
2. Flags pull_request_target with checkout of untrusted PR head.
3. Rewrites workflow to pull_request with read-only token permissions.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 246: Git Secret Scrubbing with BFG Repo-Cleaner: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Git History Engine, AgentShield, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Detect leaked AWS secret key in git history and rewrite commit tree"
```

```
```
```

- **Execution Flow**:

1. Scans commit history for high-entropy tokens.
 2. Discovers exposed secret committed in old config.
 3. Invokes BFG cleaner, forces push, and rotates IAM key.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Leaked secret purged from git history and revoked on AWS. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 247: Container SBOM Generator & Trivy CVE Matcher: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Trivy MCP, Syft SBOM Engine, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Scan container image api:v2.1, generate CycloneDX SBOM, and filter exploitable CVEs"
```

```
```
```

- **Execution Flow**:

1. Generates SBOM containing all OS and package dependencies.
 2. Cross-references NVD database.
 3. Filters out 18 CVEs in uncalled test libraries; patches 1 active CVE.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Base image upgraded; zero critical CVEs remaining. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 248: Cosign Container Image Signature & Rekor Attestation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Cosign MCP, Sigstore Rekor, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Sign container image release/v3.0 using keyless OIDC and publish Rekor attestation"
```

```
```
```

- **Execution Flow**:

1. Requests ephemeral signing certificate via GitHub Actions OIDC.

2. Signs container image digest.

3. Publishes attestation to Rekor transparency log.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Container image cryptographically verified for Kubernetes admission. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 249: Semgrep Static Analysis Rule Synthesizer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Semgrep MCP, AST Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Scan codebase for unescaped user input inside raw SQL queries and generate PR"
```

```
```
```

- **Execution Flow**:

1. Executes Semgrep AST pattern match on Python queries.

2. Discovers 4 raw format strings in database DAO.

3. Rewrites queries using SQLAlchemy parameter bindings.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: SQL injection vulnerability remediated with automated tests. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 250: GitHub Actions Workflow Hardening against PwnPR: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `GitHub Actions MCP, AgentShield, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Audit .github/workflows for dangerous pull_request_target triggers and checkout attacks"
```

```
```
```

- **Execution Flow**:

1. Parses workflow YAML abstract syntax trees.
 2. Flags pull_request_target with checkout of untrusted PR head.
 3. Rewrites workflow to pull_request with read-only token permissions.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Supply chain pipeline hardened against arbitrary code execution. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 251–300: Full-Stack Web Development, Modern Frontend & Backend APIs

Scenario 251: Next.js 15 Server Components Migration: Enterprise Baseline

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
 2. Replaces client useEffect with async Server Components.
 3. Verifies zero bundle size regression.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Production baseline operational state verified with zero drift.

Scenario 252: Database N+1 Query Elimination in Axum Rust Service: Enterprise Baseline

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Production baseline operational state verified with zero drift.

Scenario 253: Interactive Terminal UI with Ratatui & Crossterm: Enterprise Baseline

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Production baseline operational state verified with zero drift.

Scenario 254: gRPC Protobuf Contract Backward Compatibility Verifier: Enterprise Baseline

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Production baseline operational state verified with zero drift.

Scenario 255: Frontend i18n Automated Extraction and Translation: Enterprise Baseline

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with `t('key')` calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Production baseline operational state verified with zero drift.

Scenario 256: Next.js 15 Server Components Migration: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client useEffect with async Server Components.
3. Verifies zero bundle size regression.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 257: Database N+1 Query Elimination in Axum Rust Service: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 258: Interactive Terminal UI with Ratatui & Crossterm: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 259: gRPC Protobuf Contract Backward Compatibility Verifier: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 260: Frontend i18n Automated Extraction and Translation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with t('key') calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 261: Next.js 15 Server Components Migration: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client useEffect with async Server Components.
3. Verifies zero bundle size regression.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 262: Database N+1 Query Elimination in Axum Rust Service: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
 2. Rewrites query with single optimized JOIN query.
 3. Validates database index usage via EXPLAIN ANALYZE.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 263: Interactive Terminal UI with Ratatui & Crossterm: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
 2. Renders gauges for CPU/RAM and tabular pod lists.
 3. Implements non-blocking keyboard event navigation.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 264: gRPC Protobuf Contract Backward Compatibility Verifier: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 265: Frontend i18n Automated Extraction and Translation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with t('key') calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 266: Next.js 15 Server Components Migration: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client useEffect with async Server Components.
3. Verifies zero bundle size regression.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 267: Database N+1 Query Elimination in Axum Rust Service: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 268: Interactive Terminal UI with Ratatui & Crossterm: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 269: gRPC Protobuf Contract Backward Compatibility Verifier: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 270: Frontend i18n Automated Extraction and Translation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with `t('key')` calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 271: Next.js 15 Server Components Migration: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client `useEffect` with async Server Components.
3. Verifies zero bundle size regression.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 272: Database N+1 Query Elimination in Axum Rust Service: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.

2. Rewrites query with single optimized JOIN query.

3. Validates database index usage via EXPLAIN ANALYZE.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 273: Interactive Terminal UI with Ratatui & Crossterm: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.

2. Renders gauges for CPU/RAM and tabular pod lists.

3. Implements non-blocking keyboard event navigation.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 274: gRPC Protobuf Contract Backward Compatibility Verifier: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 275: Frontend i18n Automated Extraction and Translation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with t('key') calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 276: Next.js 15 Server Components Migration: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client useEffect with async Server Components.
3. Verifies zero bundle size regression.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 277: Database N+1 Query Elimination in Axum Rust Service: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 278: Interactive Terminal UI with Ratatui & Crossterm: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 279: gRPC Protobuf Contract Backward Compatibility Verifier: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 280: Frontend i18n Automated Extraction and Translation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with `t('key')` calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 281: Next.js 15 Server Components Migration: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client `useEffect` with async Server Components.
3. Verifies zero bundle size regression.
4. Activates `TAGISAN_LOCAL_ONLY=1` privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 282: Database N+1 Query Elimination in Axum Rust Service: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 283: Interactive Terminal UI with Ratatui & Crossterm: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 284: gRPC Protobuf Contract Backward Compatibility Verifier: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 285: Frontend i18n Automated Extraction and Translation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with t('key') calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 286: Next.js 15 Server Components Migration: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client useEffect with async Server Components.
3. Verifies zero bundle size regression.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 287: Database N+1 Query Elimination in Axum Rust Service: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
2. Rewrites query with single optimized JOIN query.
3. Validates database index usage via EXPLAIN ANALYZE.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 288: Interactive Terminal UI with Ratatui & Crossterm: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
2. Renders gauges for CPU/RAM and tabular pod lists.
3. Implements non-blocking keyboard event navigation.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 289: gRPC Protobuf Contract Backward Compatibility Verifier: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 290: Frontend i18n Automated Extraction and Translation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with `t('key')` calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 291: Next.js 15 Server Components Migration: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

- **Execution Flow**:

1. Separates client state hooks from pure render trees.
2. Replaces client `useEffect` with `async Server Components`.
3. Verifies zero bundle size regression.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 292: Database N+1 Query Elimination in Axum Rust Service: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

- **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.

2. Rewrites query with single optimized JOIN query.

3. Validates database index usage via EXPLAIN ANALYZE.

4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.

5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 293: Interactive Terminal UI with Ratatui & Crossterm: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.

2. Renders gauges for CPU/RAM and tabular pod lists.

3. Implements non-blocking keyboard event navigation.

4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.

5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 294: gRPC Protobuf Contract Backward Compatibility Verifier: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
2. Flags deletion of field #4 as breaking change for mobile apps.
3. Suggests reserved field tag and addition of field #5.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 295: Frontend i18n Automated Extraction and Translation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

- **Execution Flow**:

1. Scans JSX components for raw text literals.
2. Replaces literals with t('key') calls.
3. Produces high-fidelity Japanese, Spanish, and German translations.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 296: Next.js 15 Server Components Migration: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Bun Toolchain, TypeScript AST Rewriter, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Migrate client-side React SPA in src/pages to Next.js 15 App Router Server Components"
```

```
```
```

• **Execution Flow**:

1. Separates client state hooks from pure render trees.
 2. Replaces client useEffect with async Server Components.
 3. Verifies zero bundle size regression.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: First Contentful Paint improved from 2.4s to 0.3s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 297: Database N+1 Query Elimination in Axum Rust Service: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `SQL Parser, Database MCP, Gemini 3 Flash, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Profile ORM queries on /organizations endpoint and replace with JOIN FETCH"
```

```
```
```

• **Execution Flow**:

1. Detects 1 parent query followed by 850 child selects.
 2. Rewrites query with single optimized JOIN query.
 3. Validates database index usage via EXPLAIN ANALYZE.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Endpoint execution time dropped from 4,200ms to 24ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 298: Interactive Terminal UI with Ratatui & Crossterm: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Rust Compiler Engine, Ratatui, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Build interactive TUI in Rust monitoring node health with ASCII sparklines"
```

```
```
```

- **Execution Flow**:

1. Synthesizes Ratatui layout splits and event loops.
 2. Renders gauges for CPU/RAM and tabular pod lists.
 3. Implements non-blocking keyboard event navigation.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Single standalone 4.2MB binary built with zero dependencies. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 299: gRPC Protobuf Contract Backward Compatibility Verifier: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Protobuf Engine, Buf CLI MCP, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Compare updated proto/billing.proto against production schema for wire-breaking changes"
```

```
```
```

- **Execution Flow**:

1. Compiles proto definitions using Buf.
 2. Flags deletion of field #4 as breaking change for mobile apps.
 3. Suggests reserved field tag and addition of field #5.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Wire protocol backward compatibility preserved. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 300: Frontend i18n Automated Extraction and Translation: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `i18n Parser Engine, Multi-Language LLM, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Extract hardcoded UI strings into en.json and synthesize translations for ja/es/de"
```

```
```
```

• **Execution Flow**:

1. Scans JSX components for raw text literals.
 2. Replaces literals with t('key') calls.
 3. Produces high-fidelity Japanese, Spanish, and German translations.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: 84 frontend screens internationalized in 3 minutes. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 301–350: Systems Programming, Rust Asynchronous Runtime & Low-Level OS

Scenario 301: Tokio Async Task Reactor Optimization: Enterprise Baseline

• **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Production Baseline`

• **Command**:

```
```bash
```

```
tgs run "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

• **Execution Flow**:

1. Traces async tasks using Tokio console.
 2. Discovers std::fs::read stalling worker thread pool.
 3. Replaces with tokio::fs and spawn_blocking for crypto hashing.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Production baseline operational state verified with zero drift.

Scenario 302: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Enterprise Baseline

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Production baseline operational state verified with zero drift.

Scenario 303: Valgrind & ASan Memory Leak Fixer for C++ Service: Enterprise Baseline

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Production baseline operational state verified with zero drift.

Scenario 304: WebAssembly Sandbox Embedding with Wasmtime: Enterprise Baseline

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Production baseline operational state verified with zero drift.

Scenario 305: Criterion.rs Continuous Benchmarking Gatekeeper: Enterprise Baseline

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Performance regression blocked from entering release. Production baseline operational state verified with zero drift.

Scenario 306: Tokio Async Task Reactor Optimization: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 307: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 308: Valgrind & ASan Memory Leak Fixer for C++ Service: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 309: WebAssembly Sandbox Embedding with Wasmtime: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 310: Criterion.rs Continuous Benchmarking Gatekeeper: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Performance regression blocked from entering release. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 311: Tokio Async Task Reactor Optimization: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers std::fs::read stalling worker thread pool.
3. Replaces with tokio::fs and spawn_blocking for crypto hashing.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 312: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 313: Valgrind & ASan Memory Leak Fixer for C++ Service: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 314: WebAssembly Sandbox Embedding with Wasmtime: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 315: Criterion.rs Continuous Benchmarking Gatekeeper: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Performance regression blocked from entering release. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 316: Tokio Async Task Reactor Optimization: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 317: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 318: Valgrind & ASan Memory Leak Fixer for C++ Service: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 319: WebAssembly Sandbox Embedding with Wasmtime: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 320: Criterion.rs Continuous Benchmarking Gatekeeper: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Performance regression blocked from entering release. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 321: Tokio Async Task Reactor Optimization: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers `one_for_one` tree restart and heals degraded node.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 322: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 323: Valgrind & ASan Memory Leak Fixer for C++ Service: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 324: WebAssembly Sandbox Embedding with Wasmtime: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 325: Criterion.rs Continuous Benchmarking Gatekeeper: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Performance regression blocked from entering release. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 326: Tokio Async Task Reactor Optimization: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 327: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 328: Valgrind & ASan Memory Leak Fixer for C++ Service: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 329: WebAssembly Sandbox Embedding with Wasmtime: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 330: Criterion.rs Continuous Benchmarking Gatekeeper: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Performance regression blocked from entering release. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 331: Tokio Async Task Reactor Optimization: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Activates `TAGISAN_LOCAL_ONLY=1` privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 332: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
2. Adds cache-line padding (64 bytes) preventing false sharing.
3. Proves memory ordering invariants with Acquire/Release semantics.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 333: Valgrind & ASan Memory Leak Fixer for C++ Service: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using std::unique_ptr.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 334: WebAssembly Sandbox Embedding with Wasmtime: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 335: Criterion.rs Continuous Benchmarking Gatekeeper: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Performance regression blocked from entering release. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 336: Tokio Async Task Reactor Optimization: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
 2. Discovers `std::fs::read` stalling worker thread pool.
 3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 337: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
 2. Adds cache-line padding (64 bytes) preventing false sharing.
 3. Proves memory ordering invariants with Acquire/Release semantics.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 338: Valgrind & ASan Memory Leak Fixer for C++ Service: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
2. Flags heap-use-after-free on socket buffer deallocation.
3. Rewrites buffer ownership using `std::unique_ptr`.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 339: WebAssembly Sandbox Embedding with Wasmtime: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 340: Criterion.rs Continuous Benchmarking Gatekeeper: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Performance regression blocked from entering release. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 341: Tokio Async Task Reactor Optimization: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

- **Execution Flow**:

1. Traces async tasks using Tokio console.
2. Discovers `std::fs::read` stalling worker thread pool.
3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 342: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

- **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
 2. Adds cache-line padding (64 bytes) preventing false sharing.
 3. Proves memory ordering invariants with Acquire/Release semantics.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 343: Valgrind & ASan Memory Leak Fixer for C++ Service: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

- **Execution Flow**:

1. Executes packet ingestion test under ASan.
 2. Flags heap-use-after-free on socket buffer deallocation.
 3. Rewrites buffer ownership using `std::unique_ptr`.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 344: WebAssembly Sandbox Embedding with Wasmtime: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

- **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
2. Configures Wasmtime engine with fuel metering and 64MB memory.
3. Executes image filter inside safe sandbox.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 345: Criterion.rs Continuous Benchmarking Gatekeeper: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
2. Detects +14.2% regression in float parsing.
3. Identifies slower standard library parser and reverts change.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Performance regression blocked from entering release. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 346: Tokio Async Task Reactor Optimization: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Rust Compiler Engine, Tokio Console, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Profile Tokio task scheduling in src/reactor.rs and eliminate blocking I/O calls"
```

```
```
```

• **Execution Flow**:

1. Traces async tasks using Tokio console.
 2. Discovers `std::fs::read` stalling worker thread pool.
 3. Replaces with `tokio::fs` and `spawn_blocking` for crypto hashing.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Throughput increased by 420%; p99 latency dropped to 4ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 347: Lock-Free Multi-Producer Multi-Consumer Ring Buffer: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Rust Atomic Engine, Gemini 3.1 Pro Low, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Synthesize a cache-aligned lock-free MPMC ring buffer in Rust with safety proofs"
```

```
```
```

• **Execution Flow**:

1. Implements circular buffer with atomic head and tail pointers.
 2. Adds cache-line padding (64 bytes) preventing false sharing.
 3. Proves memory ordering invariants with Acquire/Release semantics.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Verified lock-free ring buffer achieving 45M ops/sec. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 348: Valgrind & ASan Memory Leak Fixer for C++ Service: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Clang/LLVM Engine, AddressSanitizer, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Compile packet parser with -fsanitize=address and isolate heap-use-after-free"
```

```
```
```

• **Execution Flow**:

1. Executes packet ingestion test under ASan.
 2. Flags heap-use-after-free on socket buffer deallocation.
 3. Rewrites buffer ownership using `std::unique_ptr`.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Memory vulnerability eliminated with zero Valgrind errors. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 349: WebAssembly Sandbox Embedding with Wasmtime: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Wasmtime Runtime Engine, Rust Toolchain, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Compile image filter in src/filters into Wasm and embed via Wasmtime engine"
```

```
```
```

• **Execution Flow**:

1. Compiles Rust to wasm32-wasi target.
 2. Configures Wasmtime engine with fuel metering and 64MB memory.
 3. Executes image filter inside safe sandbox.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Wasm execution achieved at 94% of native speeds. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 350: Criterion.rs Continuous Benchmarking Gatekeeper: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Criterion.rs Engine, GitHub Actions MCP, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Run criterion benchmark suite comparing current commit against main branch"
```

```
```
```

- **Execution Flow**:

1. Executes 10,000 iterations of serialization benchmark.
 2. Detects +14.2% regression in float parsing.
 3. Identifies slower standard library parser and reverts change.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Performance regression blocked from entering release. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 351–400: ⚡ Polyglot Concurrency, Gleam & Erlang/Elixir BEAM OTP Systems

Scenario 351: Compiling Type-Safe Gleam Microservice to BEAM: Enterprise Baseline

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
 2. Verifies exhaustive pattern matching on all domain events.
 3. Emits validated .beam bytecode for Erlang nodes.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Production baseline operational state verified with zero drift.

Scenario 352: OTP one_for_one Supervisor Crash Isolation: Enterprise Baseline

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Production baseline operational state verified with zero drift.

Scenario 353: Binary ETF (External Term Format 131) Serialization: Enterprise Baseline

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Production baseline operational state verified with zero drift.

Scenario 354: Bounded Actor Mailbox Queue with Backpressure: Enterprise Baseline

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Production baseline operational state verified with zero drift.

Scenario 355: Hot Code Reloading on Live Elixir Node: Enterprise Baseline

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via `:code.load_binary/3`.
3. Running processes transition to new code on next loop iteration.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Production baseline operational state verified with zero drift.

Scenario 356: Compiling Type-Safe Gleam Microservice to BEAM: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 357: OTP one_for_one Supervisor Crash Isolation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 358: Binary ETF (External Term Format 131) Serialization: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 359: Bounded Actor Mailbox Queue with Backpressure: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 360: Hot Code Reloading on Live Elixir Node: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via :code.load_binary/3.
3. Running processes transition to new code on next loop iteration.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 361: Compiling Type-Safe Gleam Microservice to BEAM: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 362: OTP one_for_one Supervisor Crash Isolation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
 2. BEAM supervisor catches crash and records stack trace.
 3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 363: Binary ETF (External Term Format 131) Serialization: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
 2. Encodes data using binary ETF format 131.
 3. Transmits over Unix domain socket with zero-copy decoding.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 364: Bounded Actor Mailbox Queue with Backpressure: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.

2. Routes dropped messages to persistent SQLite dead-letter queue.

3. Emits backpressure signal to upstream producers.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 365: Hot Code Reloading on Live Elixir Node: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.

2. Loads module into live runtime via :code.load_binary/3.

3. Running processes transition to new code on next loop iteration.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 366: Compiling Type-Safe Gleam Microservice to BEAM: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 367: OTP one_for_one Supervisor Crash Isolation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 368: Binary ETF (External Term Format 131) Serialization: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 369: Bounded Actor Mailbox Queue with Backpressure: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 370: Hot Code Reloading on Live Elixir Node: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via :code.load_binary/3.
3. Running processes transition to new code on next loop iteration.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 371: Compiling Type-Safe Gleam Microservice to BEAM: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 372: OTP one_for_one Supervisor Crash Isolation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
 2. BEAM supervisor catches crash and records stack trace.
 3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 373: Binary ETF (External Term Format 131) Serialization: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
 2. Encodes data using binary ETF format 131.
 3. Transmits over Unix domain socket with zero-copy decoding.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 374: Bounded Actor Mailbox Queue with Backpressure: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.

2. Routes dropped messages to persistent SQLite dead-letter queue.

3. Emits backpressure signal to upstream producers.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 375: Hot Code Reloading on Live Elixir Node: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.

2. Loads module into live runtime via :code.load_binary/3.

3. Running processes transition to new code on next loop iteration.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 376: Compiling Type-Safe Gleam Microservice to BEAM: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 377: OTP one_for_one Supervisor Crash Isolation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 378: Binary ETF (External Term Format 131) Serialization: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 379: Bounded Actor Mailbox Queue with Backpressure: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 380: Hot Code Reloading on Live Elixir Node: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via :code.load_binary/3.
3. Running processes transition to new code on next loop iteration.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 381: Compiling Type-Safe Gleam Microservice to BEAM: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 382: OTP one_for_one Supervisor Crash Isolation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 383: Binary ETF (External Term Format 131) Serialization: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 384: Bounded Actor Mailbox Queue with Backpressure: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 385: Hot Code Reloading on Live Elixir Node: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via :code.load_binary/3.
3. Running processes transition to new code on next loop iteration.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 386: Compiling Type-Safe Gleam Microservice to BEAM: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 387: OTP one_for_one Supervisor Crash Isolation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
2. BEAM supervisor catches crash and records stack trace.
3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 388: Binary ETF (External Term Format 131) Serialization: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
2. Encodes data using binary ETF format 131.
3. Transmits over Unix domain socket with zero-copy decoding.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 389: Bounded Actor Mailbox Queue with Backpressure: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
2. Routes dropped messages to persistent SQLite dead-letter queue.
3. Emits backpressure signal to upstream producers.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 390: Hot Code Reloading on Live Elixir Node: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
2. Loads module into live runtime via :code.load_binary/3.
3. Running processes transition to new code on next loop iteration.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 391: Compiling Type-Safe Gleam Microservice to BEAM: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
2. Verifies exhaustive pattern matching on all domain events.
3. Emits validated .beam bytecode for Erlang nodes.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 392: OTP one_for_one Supervisor Crash Isolation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
 2. BEAM supervisor catches crash and records stack trace.
 3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 393: Binary ETF (External Term Format 131) Serialization: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
 2. Encodes data using binary ETF format 131.
 3. Transmits over Unix domain socket with zero-copy decoding.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 394: Bounded Actor Mailbox Queue with Backpressure: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
 2. Routes dropped messages to persistent SQLite dead-letter queue.
 3. Emits backpressure signal to upstream producers.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 395: Hot Code Reloading on Live Elixir Node: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
 2. Loads module into live runtime via `:code.load_binary/3`.
 3. Running processes transition to new code on next loop iteration.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 396: Compiling Type-Safe Gleam Microservice to BEAM: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Native Gleam Compiler, BEAM VM, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

```
```
```

- **Execution Flow**:

1. Invokes native Gleam compiler.
 2. Verifies exhaustive pattern matching on all domain events.
 3. Emits validated .beam bytecode for Erlang nodes.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Type-safe BEAM bytecode produced in 420ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 397: OTP one_for_one Supervisor Crash Isolation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `OTP Supervisor Engine, Fault Simulator, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Simulate fatal panic in worker actor #4 and verify OTP supervisor auto-restart"
```

```
```
```

- **Execution Flow**:

1. Injects divide-by-zero panic in running GenServer.
 2. BEAM supervisor catches crash and records stack trace.
 3. Restarts failed worker with fresh state in 2ms without disturbing siblings.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: 99.999% uptime maintained via 'let it crash' resilience. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 398: Binary ETF (External Term Format 131) Serialization: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `ETF Codec Engine, Rust-BEAM Bridge, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Serialize 100,000 telemetry records into Erlang External Term Format (ETF) in Rust"
```

```
```
```

- **Execution Flow**:

1. Maps Rust struct hierarchy to Erlang atoms, tuples, and binaries.
 2. Encodes data using binary ETF format 131.
 3. Transmits over Unix domain socket with zero-copy decoding.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: 820,000 records/sec serialized; 40% smaller than JSON. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 399: Bounded Actor Mailbox Queue with Backpressure: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Gleam OTP Engine, Mailbox Watcher, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Implement bounded actor mailbox queue in Gleam with dead-letter queue"
```

```
```
```

- **Execution Flow**:

1. Implements actor receiver loop dropping low-priority events when queue > 10,000.
 2. Routes dropped messages to persistent SQLite dead-letter queue.
 3. Emits backpressure signal to upstream producers.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Actor process protected from OOM under 50x traffic surge. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 400: Hot Code Reloading on Live Elixir Node: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `BEAM Hot-Code Reloader, Elixir SDK, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Deploy updated payment_calc.ex to live production BEAM cluster without dropping connections"
```

```
```
```

- **Execution Flow**:

1. Compiles modified Elixir source to .beam object.
 2. Loads module into live runtime via `:code.load_binary/3`.
 3. Running processes transition to new code on next loop iteration.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Live production code hot-swapped in 15ms with 0 downtime. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 401–450: Big Data Engineering, Data Lakehouses & Real-Time Streaming

Scenario 401: Kafka Consumer Group Rebalance Minimizer: Enterprise Baseline

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies `CommitFailedException` causing rebalance storm.
 2. Increases `max.poll.interval.ms`.
 3. Upgrades partition strategy to `CooperativeStickyAssignor`.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Production baseline operational state verified with zero drift.

Scenario 402: Debezium MySQL CDC into Apache Iceberg: Enterprise Baseline

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Production baseline operational state verified with zero drift.

Scenario 403: Snowflake SQL Query Cost & Partition Pruning Optimizer: Enterprise Baseline

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Production baseline operational state verified with zero drift.

Scenario 404: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Enterprise Baseline

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Production baseline operational state verified with zero drift.

Scenario 405: Apache Spark Join Skew Salting Remediation: Enterprise Baseline

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Production baseline operational state verified with zero drift.

Scenario 406: Kafka Consumer Group Rebalance Minimizer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 407: Debezium MySQL CDC into Apache Iceberg: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 408: Snowflake SQL Query Cost & Partition Pruning Optimizer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 409: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 410: Apache Spark Join Skew Salting Remediation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 411: Kafka Consumer Group Rebalance Minimizer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 412: Debezium MySQL CDC into Apache Iceberg: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 413: Snowflake SQL Query Cost & Partition Pruning Optimizer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 414: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 415: Apache Spark Join Skew Salting Remediation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 416: Kafka Consumer Group Rebalance Minimizer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 417: Debezium MySQL CDC into Apache Iceberg: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 418: Snowflake SQL Query Cost & Partition Pruning Optimizer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 419: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 420: Apache Spark Join Skew Salting Remediation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 421: Kafka Consumer Group Rebalance Minimizer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 422: Debezium MySQL CDC into Apache Iceberg: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 423: Snowflake SQL Query Cost & Partition Pruning Optimizer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 424: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 425: Apache Spark Join Skew Salting Remediation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 426: Kafka Consumer Group Rebalance Minimizer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
 2. Increases max.poll.interval.ms.
 3. Upgrades partition strategy to CooperativeStickyAssignor.
 4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
 5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.
- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 427: Debezium MySQL CDC into Apache Iceberg: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
 2. Writes update records to partitioned Parquet files.
 3. Commits snapshot to Apache Iceberg with ACID guarantees.
 4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
 5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.
- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 428: Snowflake SQL Query Cost & Partition Pruning Optimizer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 429: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 430: Apache Spark Join Skew Salting Remediation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 431: Kafka Consumer Group Rebalance Minimizer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 432: Debezium MySQL CDC into Apache Iceberg: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 433: Snowflake SQL Query Cost & Partition Pruning Optimizer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 434: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 435: Apache Spark Join Skew Salting Remediation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 436: Kafka Consumer Group Rebalance Minimizer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 437: Debezium MySQL CDC into Apache Iceberg: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 438: Snowflake SQL Query Cost & Partition Pruning Optimizer: Predictive Telemetry & Anomaly Hunting

• **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, PILOT Memory, Predictive ML`

• **Command**:

```
```bash
```

```
tgs run --predictive-audit "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

• **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 439: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Predictive Telemetry & Anomaly Hunting

• **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, PILOT Memory, Predictive ML`

• **Command**:

```
```bash
```

```
tgs run --predictive-audit "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

• **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 440: Apache Spark Join Skew Salting Remediation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 441: Kafka Consumer Group Rebalance Minimizer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.
2. Increases max.poll.interval.ms.
3. Upgrades partition strategy to CooperativeStickyAssignor.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 442: Debezium MySQL CDC into Apache Iceberg: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.
2. Writes update records to partitioned Parquet files.
3. Commits snapshot to Apache Iceberg with ACID guarantees.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 443: Snowflake SQL Query Cost & Partition Pruning Optimizer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
2. Discovers full table scan on 2-billion-row events table.
3. Redesigns clustering key enabling 99.2% partition pruning.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 444: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
2. Executes multi-stage window aggregations across 8 CPU cores.
3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 445: Apache Spark Join Skew Salting Remediation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
2. Identifies key 'default_org' causing severe data skew.
3. Applies key salting with random integer 0..16 to distribute partitions.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 446: Kafka Consumer Group Rebalance Minimizer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Kafka Admin MCP, AgentShield, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Tune consumer group order-processing with CooperativeStickyAssignor"
```

```
```
```

- **Execution Flow**:

1. Identifies CommitFailedException causing rebalance storm.

2. Increases max.poll.interval.ms.

3. Upgrades partition strategy to CooperativeStickyAssignor.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Rebalance downtime eliminated; throughput at 85,000 msgs/s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 447: Debezium MySQL CDC into Apache Iceberg: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Debezium MCP, Iceberg Catalog Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Configure Debezium CDC pipeline streaming MySQL binlogs into Apache Iceberg table"
```

```
```
```

- **Execution Flow**:

1. Establishes Debezium connector tracking row changes.

2. Writes update records to partitioned Parquet files.

3. Commits snapshot to Apache Iceberg with ACID guarantees.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Sub-5-second lakehouse freshness achieved with zero DB load. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 448: Snowflake SQL Query Cost & Partition Pruning Optimizer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Snowflake MCP, SQL AST Optimizer, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Optimize top 10 most expensive Snowflake queries and redesign clustering keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes SNOWFLAKE.ACCOUNT_USAGE query history.
 2. Discovers full table scan on 2-billion-row events table.
 3. Redesigns clustering key enabling 99.2% partition pruning.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Query runtime reduced from 45s to 1.1s; cost cut by 60%. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 449: DuckDB In-Memory OLAP Analytics on 50GB Parquet: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `DuckDB Native Engine, Local Ollama, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Execute analytical window queries over 50GB Parquet files using DuckDB in Rust"
```

```
```
```

- **Execution Flow**:

1. Mounts Parquet directory using zero-copy reader.
 2. Executes multi-stage window aggregations across 8 CPU cores.
 3. Emits summarized JSON metrics in 1.8 seconds using < 2GB RAM.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Instant local OLAP processing achieved without cloud costs. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 450: Apache Spark Join Skew Salting Remediation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Spark Profiler MCP, Dialectical Synthesis, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Fix Spark executor OOM on stage 4 join by applying salting technique to skewed keys"
```

```
```
```

- **Execution Flow**:

1. Analyzes Spark UI; spots 1 executor processing 85% of shuffle.
 2. Identifies key 'default_org' causing severe data skew.
 3. Applies key salting with random integer 0..16 to distribute partitions.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Spark job completed in 6 minutes with zero OOM errors. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 451–500: Database Administration, SQL Optimization & Storage Engines

Scenario 451: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Enterprise Baseline

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
 2. Synthesis crafts 3-step zero-lock migration.
 3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Production baseline operational state verified with zero drift.

Scenario 452: ClickHouse SummingMergeTree Materialized View: Enterprise Baseline

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Production baseline operational state verified with zero drift.

Scenario 453: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Enterprise Baseline

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Production baseline operational state verified with zero drift.

Scenario 454: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Enterprise Baseline

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks `pg_stat_replication`; identifies I/O bottleneck on replica.
2. Increases `wal_buffers` to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Production baseline operational state verified with zero drift.

Scenario 455: ScyllaDB Wide-Column Partition Key Distribution Tuner: Enterprise Baseline

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Production baseline operational state verified with zero drift.

Scenario 456: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 457: ClickHouse SummingMergeTree Materialized View: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 458: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 459: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 460: ScyllaDB Wide-Column Partition Key Distribution Tuner: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 461: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 462: ClickHouse SummingMergeTree Materialized View: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
 2. Defines Materialized View aggregating counts on insert.
 3. Verifies dashboard query latency drops from 12s to 8ms.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 463: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
 2. Migrates hash slots to newly added node with 0 drops.
 3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 464: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
 2. Increases wal_buffers to 64MB and configures asynchronous commit.
 3. Replication lag recovers from 80GB to 0MB in 8 minutes.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 465: ScyllaDB Wide-Column Partition Key Distribution Tuner: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
 2. Adds bucket timestamp into compound partition key.
 3. Verifies smooth data distribution across all 16 cluster nodes.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 466: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 467: ClickHouse SummingMergeTree Materialized View: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 468: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 469: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 470: ScyllaDB Wide-Column Partition Key Distribution Tuner: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.

2. Adds bucket timestamp into compound partition key.

3. Verifies smooth data distribution across all 16 cluster nodes.

4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.

5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 471: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.

2. Synthesis crafts 3-step zero-lock migration.

3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 472: ClickHouse SummingMergeTree Materialized View: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
 2. Defines Materialized View aggregating counts on insert.
 3. Verifies dashboard query latency drops from 12s to 8ms.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 473: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
 2. Migrates hash slots to newly added node with 0 drops.
 3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 474: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks `pg_stat_replication`; identifies I/O bottleneck on replica.
2. Increases `wal_buffers` to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers `one_for_one` tree restart and heals degraded node.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 475: ScyllaDB Wide-Column Partition Key Distribution Tuner: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers `one_for_one` tree restart and heals degraded node.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 476: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 477: ClickHouse SummingMergeTree Materialized View: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 478: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 479: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 480: ScyllaDB Wide-Column Partition Key Distribution Tuner: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
 2. Adds bucket timestamp into compound partition key.
 3. Verifies smooth data distribution across all 16 cluster nodes.
 4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
 5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.
- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 481: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
 2. Synthesis crafts 3-step zero-lock migration.
 3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 482: ClickHouse SummingMergeTree Materialized View: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
 2. Defines Materialized View aggregating counts on insert.
 3. Verifies dashboard query latency drops from 12s to 8ms.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 483: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
 2. Migrates hash slots to newly added node with 0 drops.
 3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 484: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 485: ScyllaDB Wide-Column Partition Key Distribution Tuner: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 486: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Predictive Telemetry & Anomaly Hunting

• **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, PILOT Memory, Predictive ML`

• **Command**:

```
```bash
```

```
tgs run --predictive-audit "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

• **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 487: ClickHouse SummingMergeTree Materialized View: Predictive Telemetry & Anomaly Hunting

• **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, PILOT Memory, Predictive ML`

• **Command**:

```
```bash
```

```
tgs run --predictive-audit "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

• **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 488: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 489: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 490: ScyllaDB Wide-Column Partition Key Distribution Tuner: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 491: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

- **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
2. Synthesis crafts 3-step zero-lock migration.
3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 492: ClickHouse SummingMergeTree Materialized View: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

- **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
2. Defines Materialized View aggregating counts on insert.
3. Verifies dashboard query latency drops from 12s to 8ms.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 493: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Redis Admin MCP, AgentShield, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

- **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
2. Migrates hash slots to newly added node with 0 drops.
3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 494: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

- **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
2. Increases wal_buffers to 64MB and configures asynchronous commit.
3. Replication lag recovers from 80GB to 0MB in 8 minutes.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 495: ScyllaDB Wide-Column Partition Key Distribution Tuner: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
2. Adds bucket timestamp into compound partition key.
3. Verifies smooth data distribution across all 16 cluster nodes.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 496: Zero-Downtime PostgreSQL Schema Migration with PgBouncer: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `PostgreSQL MCP, Dialectical Debate, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Execute zero-lock schema migration on 50M-row users table adding indexed UUID column"
```

```
```
```

• **Execution Flow**:

1. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock.
 2. Synthesis crafts 3-step zero-lock migration.
 3. Executes ADD COLUMN NULLABLE -> BACKFILL -> VALIDATE CONSTRAINT.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: 50M row migration completed with 0ms query lock latency. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 497: ClickHouse SummingMergeTree Materialized View: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `ClickHouse MCP, SQL Optimizer, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API metrics"
```

```
```
```

• **Execution Flow**:

1. Creates SummingMergeTree table partitioned by month.
 2. Defines Materialized View aggregating counts on insert.
 3. Verifies dashboard query latency drops from 12s to 8ms.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Billion-row real-time analytics enabled instantly. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 498: Redis Cluster Hash Slot Rebalancer & LRU Eviction: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Redis Admin MCP, AgentShield, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Rebalance 2,048 hash slots across 6-node Redis cluster and configure volatile-lru"
```

```
```
```

• **Execution Flow**:

1. Discovers node 3 operating at 96% memory capacity.
 2. Migrates hash slots to newly added node with 0 drops.
 3. Sets maxmemory-policy volatile-lru preventing OOM crashes.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Cluster memory balanced evenly at 68% across all nodes. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 499: PostgreSQL Write-Ahead-Log (WAL) Replication Lag Healer: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Postgres Admin MCP, Linux I/O Tools, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Diagnose 80GB WAL replication lag on standby replica and tune max_parallel_workers"
```

```
```
```

• **Execution Flow**:

1. Checks pg_stat_replication; identifies I/O bottleneck on replica.
 2. Increases wal_buffers to 64MB and configures asynchronous commit.
 3. Replication lag recovers from 80GB to 0MB in 8 minutes.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Standby replica caught up to primary with zero data loss. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 500: ScyllaDB Wide-Column Partition Key Distribution Tuner: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `ScyllaDB / Cassandra MCP, NoSQL Engine, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
 tgs debate --rounds 3 "Audit ScyllaDB schema for large partitions (> 100MB) and re-partition by hour"
```

```
```
```

- **Execution Flow**:

1. Identifies oversized partition key causing hotspotting on node 2.
 2. Adds bucket timestamp into compound partition key.
 3. Verifies smooth data distribution across all 16 cluster nodes.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Hotspot latency spike eliminated; p99 write latency < 2ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 501–550: 🧠 Artificial Intelligence, Local LLMs & RAG Vector Architectures

Scenario 501: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Enterprise Baseline

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Production Baseline`
- **Command**:

```
```bash
```

```
 tgs run "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
 2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
 3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Production baseline operational state verified with zero drift.

Scenario 502: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Enterprise Baseline

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with `antigravity/2.0.0` user agent.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Production baseline operational state verified with zero drift.

Scenario 503: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Enterprise Baseline

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves `-m pro` alias to `gemini-3.1-pro-low`.
2. Model activates multi-step internal thinking chain.
3. Emits verified Rust code with atomic CAS loops.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Production baseline operational state verified with zero drift.

Scenario 504: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Enterprise Baseline

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Production baseline operational state verified with zero drift.

Scenario 505: Hegelian Dialectical Debate for AI Hallucination Elimination: Enterprise Baseline

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Production baseline operational state verified with zero drift.

Scenario 506: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tg run --failover-dr "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 507: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tg run --failover-dr "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 508: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves -m pro alias to gemini-3.1-pro-low.
2. Model activates multi-step internal thinking chain.
3. Emits verified Rust code with atomic CAS loops.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 509: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 510: Hegelian Dialectical Debate for AI Hallucination Elimination: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 511: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 512: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
 2. Validates 90s proactive token expiration cushion.
 3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 513: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves `-m pro` alias to `gemini-3.1-pro-low`.
 2. Model activates multi-step internal thinking chain.
 3. Emits verified Rust code with atomic CAS loops.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 514: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Zero-Trust Security & SOC2 Compliance

• **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, AgentShield, Zero-Trust, SOC2`

• **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

• **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
 2. Sends minimal payload directly to edge endpoint.
 3. Streams response tokens with time-to-first-token under 280ms.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 515: Hegelian Dialectical Debate for AI Hallucination Elimination: Zero-Trust Security & SOC2 Compliance

• **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, AgentShield, Zero-Trust, SOC2`

• **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

• **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
 2. Challenger proves boundary overhead and lack of vectorization in custom loops.
 3. Judge reviews cross-examination and renders binding verdict.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 516: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 517: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with `antigravity/2.0.0` user agent.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 518: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves -m pro alias to gemini-3.1-pro-low.
2. Model activates multi-step internal thinking chain.
3. Emits verified Rust code with atomic CAS loops.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 519: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 520: Hegelian Dialectical Debate for AI Hallucination Elimination: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 521: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 522: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 523: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves `-m pro` alias to `gemini-3.1-pro-low`.
2. Model activates multi-step internal thinking chain.
3. Emits verified Rust code with atomic CAS loops.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 524: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 525: Hegelian Dialectical Debate for AI Hallucination Elimination: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 526: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 527: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 528: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Multi-Tenant Enterprise Isolation

• **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Multi-Tenancy, Cryptographic Boundaries`

• **Command**:

```
```bash
```

```
tgs run --tenant-guard "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

• **Execution Flow**:

1. Resolves -m pro alias to gemini-3.1-pro-low.
 2. Model activates multi-step internal thinking chain.
 3. Emits verified Rust code with atomic CAS loops.
 4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
 5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.
- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 529: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Multi-Tenant Enterprise Isolation

• **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Multi-Tenancy, Cryptographic Boundaries`

• **Command**:

```
```bash
```

```
tgs run --tenant-guard "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

• **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
 2. Sends minimal payload directly to edge endpoint.
 3. Streams response tokens with time-to-first-token under 280ms.
 4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
 5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.
- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 530: Hegelian Dialectical Debate for AI Hallucination Elimination: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 531: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 532: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.

2. Validates 90s proactive token expiration cushion.

3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.

4. Activates `TAGISAN_LOCAL_ONLY=1` privacy lock restricting all outbound internet egress.

5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 533: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves `-m pro` alias to `gemini-3.1-pro-low`.

2. Model activates multi-step internal thinking chain.

3. Emits verified Rust code with atomic CAS loops.

4. Activates `TAGISAN_LOCAL_ONLY=1` privacy lock restricting all outbound internet egress.

5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 534: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 535: Hegelian Dialectical Debate for AI Hallucination Elimination: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 536: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 537: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini\_oauth.json`.
2. Validates 90s proactive token expiration cushion.
3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 538: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves -m pro alias to gemini-3.1-pro-low.
2. Model activates multi-step internal thinking chain.
3. Emits verified Rust code with atomic CAS loops.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 539: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 540: Hegelian Dialectical Debate for AI Hallucination Elimination: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 541: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 542: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
 2. Validates 90s proactive token expiration cushion.
 3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 543: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

- **Execution Flow**:

1. Resolves `-m pro` alias to `gemini-3.1-pro-low`.
 2. Model activates multi-step internal thinking chain.
 3. Emits verified Rust code with atomic CAS loops.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 544: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

- **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with time-to-first-token under 280ms.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 545: Hegelian Dialectical Debate for AI Hallucination Elimination: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:

1. Proposer claims NumPy matches C due to BLAS.
2. Challenger proves boundary overhead and lack of vectorization in custom loops.
3. Judge reviews cross-examination and renders binding verdict.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 546: Dual-Brain Inference Routing: Local GGUF vs. Cloud Frontier: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Tagisan Dual-Brain Router, Ollama + Gemini, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Route simple formatting queries to Ollama and complex legal reviews to Gemini Pro"
```

```
```
```

- **Execution Flow**:

1. Evaluates prompt complexity score via local classifier.
 2. Routes basic tasks to Ollama (0ms latency, \$0 cost).
 3. Fails over complex 80-page legal audit to Gemini 2.5 Pro.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: 82% of queries handled locally for free; complex tasks get frontier reasoning. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 547: Google Web OAuth CCPA Endpoint Dispatching (gemini-2.5-flash): Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Google OAuth Manager, CCPA Internal Gateway, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Explain quantum entanglement proof in 3 concise mathematical sentences"
```

```
```
```

- **Execution Flow**:

1. Verifies credentials in `~/config/tagisan/gemini_oauth.json`.
 2. Validates 90s proactive token expiration cushion.
 3. Dispatches payload to CCPA endpoint with antigravity/2.0.0 user agent.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Response received in 1.38s with zero API billing costs. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 548: Deep Reasoning Problem Solving with gemini-3.1-pro-low: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Google OAuth Endpoint, Gemini 3.1 Pro Low, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Synthesize a lock-free ring buffer algorithm in Rust with formal safety proofs"
```

```
```
```

• **Execution Flow**:

1. Resolves -m pro alias to gemini-3.1-pro-low.

2. Model activates multi-step internal thinking chain.

3. Emits verified Rust code with atomic CAS loops.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: High-complexity algorithm solved with formal reasoning in 3.7s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 549: Ultra-Low Latency Inline Assistant with gemini-2.5-flash-lite: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Google OAuth Endpoint, Gemini 2.5 Flash Lite, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Generate 5 high-speed Linux sysadmin shorthand aliases for network troubleshooting"
```

```
```
```

• **Execution Flow**:

1. Resolves -m lite alias to gemini-2.5-flash-lite.

2. Sends minimal payload directly to edge endpoint.

3. Streams response tokens with time-to-first-token under 280ms.

4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.

5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Instantaneous completion received in 1.02s. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 550: Hegelian Dialectical Debate for AI Hallucination Elimination: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Swarm MoA Debate Engine, 4-Agent Consensus, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Resolve dispute: Python with NumPy is fundamentally faster than native Rust loops"
```

```
```
```

- **Execution Flow**:
 1. Proposer claims NumPy matches C due to BLAS.
 2. Challenger proves boundary overhead and lack of vectorization in custom loops.
 3. Judge reviews cross-examination and renders binding verdict.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Factually verified consensus synthesized with zero hallucinations. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 551–600: Machine Learning Engineering, MLOps & Model Serving

Scenario 551: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Enterprise Baseline

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Production Baseline`
- **Command**:

```
```bash
```

```
tgs run "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:
 1. Converts Safetensors weights to FP16 GGUF.
 2. Executes llama-quantize with Q4_K_M matrix.
 3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
 4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Production baseline operational state verified with zero drift.

Scenario 552: LoRA Fine-Tuning on Domain APIs with Unsloth: Enterprise Baseline

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Production baseline operational state verified with zero drift.

Scenario 553: Qdrant HNSW Vector Index Tuning (< 10ms Search): Enterprise Baseline

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Production baseline operational state verified with zero drift.

Scenario 554: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Enterprise Baseline

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Production baseline operational state verified with zero drift.

Scenario 555: Adversarial Prompt Injection Defense Benchmark: Enterprise Baseline

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Production baseline operational state verified with zero drift.

Scenario 556: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 557: LoRA Fine-Tuning on Domain APIs with Unsloth: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 558: Qdrant HNSW Vector Index Tuning (< 10ms Search): Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.

2. Validates 98.6% recall maintained.

3. Memory footprint reduced by 75%.

4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.

5. Executes automated health check cutover validation.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 559: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.

2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).

3. Reranks top 20 candidates; returns top 3 precision passages.

4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.

5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 560: Adversarial Prompt Injection Defense Benchmark: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 561: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 562: LoRA Fine-Tuning on Domain APIs with Unsloth: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.

2. Injects trainable LoRA matrices into attention layers.

3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 563: Qdrant HNSW Vector Index Tuning (< 10ms Search): Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.

2. Validates 98.6% recall maintained.

3. Memory footprint reduced by 75%.

4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.

5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 564: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Zero-Trust Security & SOC2 Compliance

• **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, AgentShield, Zero-Trust, SOC2`

• **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

• **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
 2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
 3. Reranks top 20 candidates; returns top 3 precision passages.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 565: Adversarial Prompt Injection Defense Benchmark: Zero-Trust Security & SOC2 Compliance

• **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, AgentShield, Zero-Trust, SOC2`

• **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

• **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
 2. AgentShield AST scanner intercepts system prompt override attempts.
 3. Intercepts hidden shell execution attempts in Markdown links.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 566: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 567: LoRA Fine-Tuning on Domain APIs with Unsloth: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 568: Qdrant HNSW Vector Index Tuning (< 10ms Search): Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 569: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 570: Adversarial Prompt Injection Defense Benchmark: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.

2. AgentShield AST scanner intercepts system prompt override attempts.

3. Intercepts hidden shell execution attempts in Markdown links.

4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.

5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 571: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.

2. Executes llama-quantize with Q4_K_M matrix.

3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.

4. Simulates random SIGKILL process termination and synthetic network partitions.

5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 572: LoRA Fine-Tuning on Domain APIs with Unsloth: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 573: Qdrant HNSW Vector Index Tuning (< 10ms Search): Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 574: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 575: Adversarial Prompt Injection Defense Benchmark: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 576: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 577: LoRA Fine-Tuning on Domain APIs with Unsloth: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 578: Qdrant HNSW Vector Index Tuning (< 10ms Search): Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash

tgs run --tenant-guard "Tune HNSW index parameters on 10M vector collection for < 10ms latency"

```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 579: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash

tgs run --tenant-guard "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"

```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 580: Adversarial Prompt Injection Defense Benchmark: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 581: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 582: LoRA Fine-Tuning on Domain APIs with Unsloth: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 583: Qdrant HNSW Vector Index Tuning (< 10ms Search): Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 584: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 585: Adversarial Prompt Injection Defense Benchmark: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 586: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 587: LoRA Fine-Tuning on Domain APIs with Unsloth: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 588: Qdrant HNSW Vector Index Tuning (< 10ms Search): Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
 2. Validates 98.6% recall maintained.
 3. Memory footprint reduced by 75%.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 589: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
 2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
 3. Reranks top 20 candidates; returns top 3 precision passages.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 590: Adversarial Prompt Injection Defense Benchmark: Predictive Telemetry & Anomaly Hunting

• **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, PILOT Memory, Predictive ML`

• **Command**:

```
```bash
```

```
tgs run --predictive-audit "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

• **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 591: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Cross-Cloud Wire Protocol Bridge

• **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Polyglot Bridge, ETF 131, Protobuf`

• **Command**:

```
```bash
```

```
tgs run --wire-bridge "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

• **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
2. Executes llama-quantize with Q4_K_M matrix.
3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 592: LoRA Fine-Tuning on Domain APIs with Unsloth: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
2. Injects trainable LoRA matrices into attention layers.
3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 593: Qdrant HNSW Vector Index Tuning (< 10ms Search): Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
2. Validates 98.6% recall maintained.
3. Memory footprint reduced by 75%.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 594: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
3. Reranks top 20 candidates; returns top 3 precision passages.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 595: Adversarial Prompt Injection Defense Benchmark: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:

1. Dispatches automated battery of prompt injection payloads.
2. AgentShield AST scanner intercepts system prompt override attempts.
3. Intercepts hidden shell execution attempts in Markdown links.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 596: Quantizing PyTorch Weights into 4-bit GGUF via llama.cpp: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `llama.cpp Toolchain, AgentShield Sandbox, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Quantize raw FP16 PyTorch model to Q4_K_M GGUF format for edge inference"
```

```
```
```

- **Execution Flow**:

1. Converts Safetensors weights to FP16 GGUF.
 2. Executes llama-quantize with Q4_K_M matrix.
 3. Verifies perplexity degradation is < 0.05% while shrinking model from 14GB to 4.2GB.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Model runs on consumer 8GB VRAM GPU at 68 tokens/sec. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 597: LoRA Fine-Tuning on Domain APIs with Unsloth: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `PyTorch / Unsloth MCP, Python uv Toolchain, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Fine-tune Qwen-2.5-Coder on 5,000 internal API examples using LoRA rank 16"
```

```
```
```

- **Execution Flow**:

1. Tokenizes domain dataset with ChatML template.
 2. Injects trainable LoRA matrices into attention layers.
 3. Completes 3 training epochs in 45 minutes; merges weights into GGUF.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Domain model achieves 99.4% accuracy on internal APIs. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 598: Qdrant HNSW Vector Index Tuning (< 10ms Search): Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Qdrant Admin MCP, Vector Benchmark Engine, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Tune HNSW index parameters on 10M vector collection for < 10ms latency"
```

```
```
```

- **Execution Flow**:

1. Reconfigures collection to m=32, ef_construct=256, scalar int8 quantization.
 2. Validates 98.6% recall maintained.
 3. Memory footprint reduced by 75%.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: P99 vector search latency clocked at 7.4 milliseconds. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 599: Hybrid Sparse/Dense RAG Search with Reciprocal Rank Fusion: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Qdrant Vector MCP, BM25 Tokenizer, RRF, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Implement hybrid RAG combining BM25 keyword matching with dense embeddings"
```

```
```
```

- **Execution Flow**:

1. Computes sparse lexical tokens and dense 1024-dim vectors in parallel.
 2. Queries Qdrant using Reciprocal Rank Fusion (RRF k=60).
 3. Reranks top 20 candidates; returns top 3 precision passages.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Retrieval MRR@10 increased from 0.71 to 0.94. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 600: Adversarial Prompt Injection Defense Benchmark: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `AgentShield Threat Evaluator, Red-Team Suite, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Execute 500 adversarial jailbreak prompts (DAN, Base64, Unicode) against AgentShield"
```

```
```
```

- **Execution Flow**:
 1. Dispatches automated battery of prompt injection payloads.
 2. AgentShield AST scanner intercepts system prompt override attempts.
 3. Intercepts hidden shell execution attempts in Markdown links.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
 - **Sovereign Outcome**: 100% of critical jailbreak payloads intercepted cleanly. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 601–650: Quantitative Finance, Algorithmic Trading & Risk (VELLA)

Scenario 601: High-Frequency Forex Tick Spread & Margin Analysis: Enterprise Baseline

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Production Baseline`
- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:
 1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
 2. Computes pip value (\$100.00/pip) across 3 broker feeds.
 3. Warns of anomalous spread widening prior to NFP release.
 4. Establishes deterministic configuration baseline and audit trail.
 - **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Production baseline operational state verified with zero drift.

Scenario 602: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Enterprise Baseline

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 correlated Gaussian paths across CPU threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Production baseline operational state verified with zero drift.

Scenario 603: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Enterprise Baseline

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
2. Detects sudden flash drop in JPY positions dropping margin to 118%.
3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Production baseline operational state verified with zero drift.

Scenario 604: Cross-Exchange Crypto Arbitrage with Gas Estimation: Enterprise Baseline

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
3. Confirms net profit of \$1,840; submits Flashbots private bundle.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Production baseline operational state verified with zero drift.

Scenario 605: Order Book Imbalance (OBI) High-Frequency Forecasting: Enterprise Baseline

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Production baseline operational state verified with zero drift.

Scenario 606: High-Frequency Forex Tick Spread & Margin Analysis: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 607: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 correlated Gaussian paths across CPU threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 608: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
2. Detects sudden flash drop in JPY positions dropping margin to 118%.
3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 609: Cross-Exchange Crypto Arbitrage with Gas Estimation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
3. Confirms net profit of \$1,840; submits Flashbots private bundle.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 610: Order Book Imbalance (OBI) High-Frequency Forecasting: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
 tgs run --failover-dr "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 611: High-Frequency Forex Tick Spread & Margin Analysis: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
 tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 612: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
 2. Generates 100,000 correlated Gaussian paths across CPU threads.
 3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 613: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
 2. Detects sudden flash drop in JPY positions dropping margin to 118%.
 3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 614: Cross-Exchange Crypto Arbitrage with Gas Estimation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
 2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
 3. Confirms net profit of \$1,840; submits Flashbots private bundle.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 615: Order Book Imbalance (OBI) High-Frequency Forecasting: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
 2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
 3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 616: High-Frequency Forex Tick Spread & Margin Analysis: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 617: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 correlated Gaussian paths across CPU threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 618: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
2. Detects sudden flash drop in JPY positions dropping margin to 118%.
3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 619: Cross-Exchange Crypto Arbitrage with Gas Estimation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
3. Confirms net profit of \$1,840; submits Flashbots private bundle.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 620: Order Book Imbalance (OBI) High-Frequency Forecasting: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 621: High-Frequency Forex Tick Spread & Margin Analysis: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 622: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
 2. Generates 100,000 correlated Gaussian paths across CPU threads.
 3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 623: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
 2. Detects sudden flash drop in JPY positions dropping margin to 118%.
 3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
 4. Simulates random SIGKILL process termination and synthetic network partitions.
 5. OTP supervisor triggers one_for_one tree restart and heals degraded node.
- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 624: Cross-Exchange Crypto Arbitrage with Gas Estimation: Chaos Resilience & Self-Healing

• **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, BEAM Supervisor, Chaos Resilience`

• **Command**:

```
```bash
```

```
tgs run --chaos-test "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

• **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
3. Confirms net profit of \$1,840; submits Flashbots private bundle.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 625: Order Book Imbalance (OBI) High-Frequency Forecasting: Chaos Resilience & Self-Healing

• **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, BEAM Supervisor, Chaos Resilience`

• **Command**:

```
```bash
```

```
tgs run --chaos-test "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

• **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 626: High-Frequency Forex Tick Spread & Margin Analysis: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 627: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 correlated Gaussian paths across CPU threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 628: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
2. Detects sudden flash drop in JPY positions dropping margin to 118%.
3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 629: Cross-Exchange Crypto Arbitrage with Gas Estimation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
3. Confirms net profit of \$1,840; submits Flashbots private bundle.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 630: Order Book Imbalance (OBI) High-Frequency Forecasting: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 631: High-Frequency Forex Tick Spread & Margin Analysis: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 632: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:
 1. Ingests covariance matrix for 20 asset classes.
 2. Generates 100,000 correlated Gaussian paths across CPU threads.
 3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 633: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:
 1. Polls equity and margin balance every 500ms.
 2. Detects sudden flash drop in JPY positions dropping margin to 118%.
 3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 634: Cross-Exchange Crypto Arbitrage with Gas Estimation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
 2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
 3. Confirms net profit of \$1,840; submits Flashbots private bundle.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 635: Order Book Imbalance (OBI) High-Frequency Forecasting: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
 2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
 3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
 4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
 5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).
- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 636: High-Frequency Forex Tick Spread & Margin Analysis: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 637: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 correlated Gaussian paths across CPU threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 638: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.

2. Detects sudden flash drop in JPY positions dropping margin to 118%.

3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.

4. Aggregates timeseries metrics across episodic memory into sliding window vectors.

5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 639: Cross-Exchange Crypto Arbitrage with Gas Estimation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.

2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).

3. Confirms net profit of \$1,840; submits Flashbots private bundle.

4. Aggregates timeseries metrics across episodic memory into sliding window vectors.

5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 640: Order Book Imbalance (OBI) High-Frequency Forecasting: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgts run --predictive-audit "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 641: High-Frequency Forex Tick Spread & Margin Analysis: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgts vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00/pip) across 3 broker feeds.
3. Warns of anomalous spread widening prior to NFP release.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 642: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
 2. Generates 100,000 correlated Gaussian paths across CPU threads.
 3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 643: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

- **Execution Flow**:

1. Polls equity and margin balance every 500ms.
 2. Detects sudden flash drop in JPY positions dropping margin to 118%.
 3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 644: Cross-Exchange Crypto Arbitrage with Gas Estimation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

- **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
 2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
 3. Confirms net profit of \$1,840; submits Flashbots private bundle.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 645: Order Book Imbalance (OBI) High-Frequency Forecasting: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
 2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
 3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
 4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
 5. Streams binary records across inter-cloud socket buffer at wire speed.
- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 646: High-Frequency Forex Tick Spread & Margin Analysis: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA Quant Engine, FIX Protocol Parser, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs vella forex --pair EUR/USD --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0
```

```
```
```

- **Execution Flow**:

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
 2. Computes pip value (\$100.00/pip) across 3 broker feeds.
 3. Warns of anomalous spread widening prior to NFP release.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Execution routed to tightest ECN provider, saving \$450 in slippage. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 647: Monte Carlo 100,000-Path Value-at-Risk (VaR 99%): Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA Monte Carlo Simulator, Rayon Threads, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Execute 100,000 Monte Carlo paths for $5M portfolio and compute 99% VaR"
```

```
```
```

- **Execution Flow**:

1. Ingests covariance matrix for 20 asset classes.
 2. Generates 100,000 correlated Gaussian paths across CPU threads.
 3. Computes 99% 10-day Value-at-Risk (\$318,400) and Expected Shortfall.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Risk report signed and submitted to Chief Risk Officer. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 648: Real-Time Margin Utilization & Pre-Liquidation De-leveraging: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `VELLA Risk Controller, Exchange REST API, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Monitor margin level; if margin level drops < 120%, close lowest conviction trade"
```

```
```
```

• **Execution Flow**:

1. Polls equity and margin balance every 500ms.
 2. Detects sudden flash drop in JPY positions dropping margin to 118%.
 3. Submits limit order closing 2 lots of USD/JPY, restoring margin to 164%.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Catastrophic account liquidation prevented automatically. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 649: Cross-Exchange Crypto Arbitrage with Gas Estimation: Hegelian Multi-Agent Formal Verification

• **TGS Capabilities**: `Web3 MCP, DEX Liquidity Math Engine, Swarm MoA, Dialectical Debate, Judge`

• **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Scan Uniswap v3 and Binance ETH/USDT price divergence and estimate net profit"
```

```
```
```

• **Execution Flow**:

1. Detects 0.65% price discrepancy between Binance spot and Uniswap pool.
 2. Computes mainnet gas fee (32 Gwei) and swap fee (0.05%).
 3. Confirms net profit of \$1,840; submits Flashbots private bundle.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Arbitrage executed on-chain without MEV sandwiching. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 650: Order Book Imbalance (OBI) High-Frequency Forecasting: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `L2/L3 Orderbook Engine, Rust AVX-512, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook"
```

```
```
```

- **Execution Flow**:

1. Ingests live WebSocket L2 orderbook updates.
 2. Computes weighted depth imbalance $(V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$.
 3. Detects institutional spoof wall pulling liquidity; alerts trading desk.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: High-frequency trade signals generated with sub-ms latency. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 651–700: Industrial IoT, SCADA Systems & Smart Infrastructure (VELLA)

Scenario 651: Modbus TCP Pressure Relief Valve Telemetry Sync: Enterprise Baseline

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Production Baseline`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
 2. Compares against safety threshold (80.0 PSI).
 3. Automatically triggers emergency cooling auxiliary pump.
 4. Establishes deterministic configuration baseline and audit trail.
- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Production baseline operational state verified with zero drift.

Scenario 652: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Enterprise Baseline

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Production baseline operational state verified with zero drift.

Scenario 653: CNC Mill Thermal Expansion Digital Twin Compensation: Enterprise Baseline

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. Production baseline operational state verified with zero drift.

Scenario 654: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Enterprise Baseline

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Production baseline operational state verified with zero drift.

Scenario 655: Allen-Bradley ControlLogix PLC Memory Mirroring: Enterprise Baseline

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Production baseline operational state verified with zero drift.

Scenario 656: Modbus TCP Pressure Relief Valve Telemetry Sync: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 657: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 658: CNC Mill Thermal Expansion Digital Twin Compensation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 659: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 660: Allen-Bradley ControlLogix PLC Memory Mirroring: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 661: Modbus TCP Pressure Relief Valve Telemetry Sync: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 662: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
 2. Flags temperature rising while cooling valve reports 100% open.
 3. Diagnoses physical valve mechanical seizure; dispatches work order.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 663: CNC Mill Thermal Expansion Digital Twin Compensation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
 2. Predicts 18.4um axial thermal expansion along Z-axis.
 3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Machining tolerance held within +-2um across 4-hour production run. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 664: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 665: Allen-Bradley ControlLogix PLC Memory Mirroring: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 666: Modbus TCP Pressure Relief Valve Telemetry Sync: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 667: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 668: CNC Mill Thermal Expansion Digital Twin Compensation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 669: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 670: Allen-Bradley ControlLogix PLC Memory Mirroring: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 671: Modbus TCP Pressure Relief Valve Telemetry Sync: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 672: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 673: CNC Mill Thermal Expansion Digital Twin Compensation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 674: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgts run --chaos-test "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 675: Allen-Bradley ControlLogix PLC Memory Mirroring: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgts run --chaos-test "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 676: Modbus TCP Pressure Relief Valve Telemetry Sync: Multi-Tenant Enterprise Isolation

• **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Multi-Tenancy, Cryptographic Boundaries`

• **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

• **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 677: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Multi-Tenant Enterprise Isolation

• **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Multi-Tenancy, Cryptographic Boundaries`

• **Command**:

```
```bash
```

```
tgs run --tenant-guard "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

• **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 678: CNC Mill Thermal Expansion Digital Twin Compensation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 679: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 680: Allen-Bradley ControlLogix PLC Memory Mirroring: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 681: Modbus TCP Pressure Relief Valve Telemetry Sync: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Air-Gap, Local Ollama, Privacy Lock`
- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 682: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 683: CNC Mill Thermal Expansion Digital Twin Compensation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Machining tolerance held within +-2um across 4-hour production run. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 684: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 685: Allen-Bradley ControlLogix PLC Memory Mirroring: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 686: Modbus TCP Pressure Relief Valve Telemetry Sync: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 687: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 688: CNC Mill Thermal Expansion Digital Twin Compensation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
 2. Predicts 18.4um axial thermal expansion along Z-axis.
 3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 689: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
 2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
 3. Estimates remaining useful life at 120 operating hours.
 4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
 5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.
- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 690: Allen-Bradley ControlLogix PLC Memory Mirroring: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgts run --predictive-audit "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 691: Modbus TCP Pressure Relief Valve Telemetry Sync: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgts vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
2. Compares against safety threshold (80.0 PSI).
3. Automatically triggers emergency cooling auxiliary pump.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 692: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
2. Flags temperature rising while cooling valve reports 100% open.
3. Diagnoses physical valve mechanical seizure; dispatches work order.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 693: CNC Mill Thermal Expansion Digital Twin Compensation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
2. Predicts 18.4um axial thermal expansion along Z-axis.
3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Machining tolerance held within +-2um across 4-hour production run. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 694: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
3. Estimates remaining useful life at 120 operating hours.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 695: Allen-Bradley ControlLogix PLC Memory Mirroring: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:

1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
2. Stores state transitions in local high-speed circular memory buffer.
3. Detects asynchronous interlock race condition between conveyor and robot.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 696: Modbus TCP Pressure Relief Valve Telemetry Sync: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA SCADA Engine, Modbus Protocol, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs vella scada --endpoint tcp://192.168.1.100:502 --analog 85.4 --alarm trip_cooling
```

```
```
```

- **Execution Flow**:

1. Connects to industrial PLC; polls holding registers for pressure (85.4 PSI).
 2. Compares against safety threshold (80.0 PSI).
 3. Automatically triggers emergency cooling auxiliary pump.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Pressure normalized back to 72.0 PSI; explosion risk prevented. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 697: OPC-UA Refinery Sensor Correlation Breakdown Anomaly: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `OPC-UA Client MCP, Anomaly Model, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Subscribe to 500 OPC-UA sensor nodes and detect correlation breakdown"
```

```
```
```

- **Execution Flow**:

1. Subscribes to live temperature, pressure, flow telemetry.
 2. Flags temperature rising while cooling valve reports 100% open.
 3. Diagnoses physical valve mechanical seizure; dispatches work order.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Faulty valve identified before catalyst bed degradation occurred. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 698: CNC Mill Thermal Expansion Digital Twin Compensation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA Digital Twin Physics Engine, C++ Math, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Simulate spindle thermal expansion on 5-axis CNC mill at 18,000 RPM for 4h"
```

```
```
```

- **Execution Flow**:

1. Solves thermal diffusion differential equations across bearings.
 2. Predicts 18.4um axial thermal expansion along Z-axis.
 3. Transmits dynamic G-code tool-length offset compensation to CNC controller.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Machining tolerance held within $\pm 2\mu\text{m}$ across 4-hour production run. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 699: Bearing Vibration FFT Spectral Analysis (Predictive Maint): Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Fast Fourier Transform (FFT) Engine, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Compute 4,096-point FFT on accelerometer timeseries from turbine generator"
```

```
```
```

- **Execution Flow**:

1. Converts 10 kHz vibration timeseries to frequency domain.
 2. Identifies sharp spectral peak at 148 Hz matching BPFO bearing frequency.
 3. Estimates remaining useful life at 120 operating hours.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Replacement scheduled during routine downtime, avoiding turbine failure. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 700: Allen-Bradley ControlLogix PLC Memory Mirroring: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `EtherNet/IP CIP Protocol Engine, VELLA Twin, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Mirror live Allen-Bradley ControlLogix PLC memory tags into SQLite digital twin"
```

```
```
```

- **Execution Flow**:
 1. Establishes EtherNet/IP CIP session polling 1,200 tags every 50ms.
 2. Stores state transitions in local high-speed circular memory buffer.
 3. Detects asynchronous interlock race condition between conveyor and robot.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Interlock bug diagnosed and patched in ladder logic in 15 minutes. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 701–750: 🚀 Aerospace Orbitals, Bioinformatics & Web3 Digital Twins (VELLA)

Scenario 701: LEO Satellite SGP4 TLE Orbit Propagation: Enterprise Baseline

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Production Baseline`
- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:
 1. Parses NORAD Two-Line Element (TLE) for ISS.
 2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
 3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
 4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Production baseline operational state verified with zero drift.

Scenario 702: Ground Station Pass Visibility & Tracking Angle Forecast: Enterprise Baseline

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Production baseline operational state verified with zero drift.

Scenario 703: Satellite Space Debris Collision Avoidance Maneuver: Enterprise Baseline

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Production baseline operational state verified with zero drift.

Scenario 704: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Enterprise Baseline

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Production Baseline`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Production baseline operational state verified with zero drift.

Scenario 705: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Enterprise Baseline

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Production Baseline`

- **Command**:

```
```bash
```

```
tgs run "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Establishes deterministic configuration baseline and audit trail.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Production baseline operational state verified with zero drift.

Scenario 706: LEO Satellite SGP4 TLE Orbit Propagation: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 707: Ground Station Pass Visibility & Tracking Angle Forecast: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 708: Satellite Space Debris Collision Avoidance Maneuver: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 709: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 710: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Multi-Region HA & Disaster Recovery

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, High-Availability, Multi-Region`

- **Command**:

```
```bash
```

```
tgs run --failover-dr "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Probes secondary region standby health and verifies cross-region replication lag < 500ms.
5. Executes automated health check cutover validation.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Active-active disaster recovery cutover validated with sub-minute RTO.

Scenario 711: LEO Satellite SGP4 TLE Orbit Propagation: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 712: Ground Station Pass Visibility & Tracking Angle Forecast: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, AgentShield, Zero-Trust, SOC2`
- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
 2. Filters passes with elevation angle > 10 degrees above horizon.
 3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 713: Satellite Space Debris Collision Avoidance Maneuver: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, AgentShield, Zero-Trust, SOC2`
- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
 2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
 3. Re-propagates orbits confirming miss distance increases to 4.8 km.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 714: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
 2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
 3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 715: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Zero-Trust Security & SOC2 Compliance

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, AgentShield, Zero-Trust, SOC2`

- **Command**:

```
```bash
```

```
tgs run --hardened --sandbox strict "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
 2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
 3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
 4. AgentShield AST scanner audits all proposed operations against NIST SP 800-53 controls.
 5. Linux Landlock LSM confines execution to isolated ephemeral scratch workspace.
- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Zero-trust compliance guardrails enforced; zero unverified privileges granted.

Scenario 716: LEO Satellite SGP4 TLE Orbit Propagation: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Tokio Reactor, P99 Optimization`
- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 717: Ground Station Pass Visibility & Tracking Angle Forecast: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Tokio Reactor, P99 Optimization`
- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 718: Satellite Space Debris Collision Avoidance Maneuver: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 719: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 720: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Sub-Millisecond P99 Performance Tuning

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Tokio Reactor, P99 Optimization`

- **Command**:

```
```bash
```

```
tgs run --opt-level 3 "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Evaluates CPU cycle latency and eliminates lock contention in worker threads.
5. Benchmarks memory throughput across SIMD AVX-512 register lanes.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. P99 latency slashed by over 80% with zero throughput degradation.

Scenario 721: LEO Satellite SGP4 TLE Orbit Propagation: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 722: Ground Station Pass Visibility & Tracking Angle Forecast: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 723: Satellite Space Debris Collision Avoidance Maneuver: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, BEAM Supervisor, Chaos Resilience`
- **Command**:

```
```bash
```

```
tgs run --chaos-test "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 724: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 725: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Chaos Resilience & Self-Healing

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, BEAM Supervisor, Chaos Resilience`

- **Command**:

```
```bash
```

```
tgs run --chaos-test "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Simulates random SIGKILL process termination and synthetic network partitions.
5. OTP supervisor triggers one_for_one tree restart and heals degraded node.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. System demonstrated 100% self-healing recovery with zero dropped user requests.

Scenario 726: LEO Satellite SGP4 TLE Orbit Propagation: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 727: Ground Station Pass Visibility & Tracking Angle Forecast: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 728: Satellite Space Debris Collision Avoidance Maneuver: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs run --tenant-guard "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 729: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Multi-Tenancy, Cryptographic Boundaries`
- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 730: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Multi-Tenant Enterprise Isolation

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Multi-Tenancy, Cryptographic Boundaries`

- **Command**:

```
```bash
```

```
 tgs run --tenant-guard "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Enforces strict tenant cryptographic boundaries and verifies row-level security (RLS).
5. Prevents noisy-neighbor noisy I/O throttling via cgroup v2 limits.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Strict cryptographic tenant isolation verified across all compute and storage layers.

Scenario 731: LEO Satellite SGP4 TLE Orbit Propagation: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
 tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 732: Ground Station Pass Visibility & Tracking Angle Forecast: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Calculate next 24-hour pass windows and Az/El tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 733: Satellite Space Debris Collision Avoidance Maneuver: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
tgs run --offline --local-only "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 734: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
 tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 735: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Air-Gapped Sovereign Local Operation

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Air-Gap, Local Ollama, Privacy Lock`

- **Command**:

```
```bash
```

```
 tgs run --offline --local-only "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Activates TAGISAN_LOCAL_ONLY=1 privacy lock restricting all outbound internet egress.
5. Dispatches inference directly to local quantized Ollama model (qwen2.5-coder / smollm2).

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Mission accomplished with 100% offline sovereignty; zero network bytes leaked.

Scenario 736: LEO Satellite SGP4 TLE Orbit Propagation: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 737: Ground Station Pass Visibility & Tracking Angle Forecast: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 738: Satellite Space Debris Collision Avoidance Maneuver: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 739: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, PILOT Memory, Predictive ML`
- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 740: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Predictive Telemetry & Anomaly Hunting

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, PILOT Memory, Predictive ML`

- **Command**:

```
```bash
```

```
tgs run --predictive-audit "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Aggregates timeseries metrics across episodic memory into sliding window vectors.
5. Forecasts metric trajectory 4 hours ahead and flags statistical standard-deviation breakout.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Impending outage predicted and remediated 3.5 hours before business impact.

Scenario 741: LEO Satellite SGP4 TLE Orbit Propagation: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 742: Ground Station Pass Visibility & Tracking Angle Forecast: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Calculate next 24-hour pass windows and Az/El tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
2. Filters passes with elevation angle > 10 degrees above horizon.
3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 743: Satellite Space Debris Collision Avoidance Maneuver: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Polyglot Bridge, ETF 131, Protobuf`
- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
3. Re-propagates orbits confirming miss distance increases to 4.8 km.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 744: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 745: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Cross-Cloud Wire Protocol Bridge

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Polyglot Bridge, ETF 131, Protobuf`

- **Command**:

```
```bash
```

```
tgs run --wire-bridge "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Compiles high-speed wire serializer converting JSON payloads to Erlang ETF format 131.
5. Streams binary records across inter-cloud socket buffer at wire speed.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Heterogeneous cross-cloud communication achieved at 850,000 messages/second.

Scenario 746: LEO Satellite SGP4 TLE Orbit Propagation: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA Aerospace Engine, SGP4 Orbit Solver, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs vella aerospace --minutes 90.0 --tle '1 25544U 98067A 26258.51460395
```

```
```
```

- **Execution Flow**:

1. Parses NORAD Two-Line Element (TLE) for ISS.
 2. Executes SGP4 perturbation model accounting for Earth oblateness (J2, J3, J4).
 3. Computes ECI state vectors (X, Y, Z) and ground track latitude/longitude.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Orbit propagated with sub-meter numerical precision. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 747: Ground Station Pass Visibility & Tracking Angle Forecast: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Orbital Geometry Engine, Ground Station MCP, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Calculate next 24-hour pass windows and Az/EI tracking angles for Svalbard station"
```

```
```
```

- **Execution Flow**:

1. Evaluates satellite position relative to Svalbard ground station.
 2. Filters passes with elevation angle > 10 degrees above horizon.
 3. Generates 6 daily pass schedules with AOS, Max Elevation, and LOS.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Ground station tracking angles exported to auto-tracker. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 748: Satellite Space Debris Collision Avoidance Maneuver: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `Conjunction Assessment Engine, Swarm MoA, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Analyze Space-Track CDM; miss distance is 142m against orbital debris"
```

```
```
```

- **Execution Flow**:

1. Ingests CDM covariance ellipsoids; calculates collision probability ($P_c = 4.8e-3$).
 2. Formulates impulsive Delta-V burn vector: 0.18 m/s along velocity vector.
 3. Re-propagates orbits confirming miss distance increases to 4.8 km.
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Thruster burn sequence approved and scheduled on next pass. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 749: FASTA Smith-Waterman Sequence Alignment & SNP Identifier: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `VELLA Bio Engine, SIMD Dynamic Programming, Swarm MoA, Dialectical Debate, Judge`
- **Command**:

```
```bash
```

```
tgs vella bio --target ACTGATCGATCGATCG --template ACTGATCGTTCGATCG --ref-genome GRCh38
```

```
```
```

- **Execution Flow**:

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
 2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
 3. Identifies single nucleotide polymorphism (SNP) at pos 9: Cytosine -> Thymine (C>T).
 4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
 5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.
- **Sovereign Outcome**: Exact alignment coordinates and substitution identified in 1.4ms. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Scenario 750: CRISPR-Cas9 On-Target and Off-Target Cleavage Scorer: Hegelian Multi-Agent Formal Verification

- **TGS Capabilities**: `CRISPR Guide RNA Engine, ML Scorer, Swarm MoA, Dialectical Debate, Judge`

- **Command**:

```
```bash
```

```
tgs debate --rounds 3 "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target scores"
```

```
```
```

- **Execution Flow**:

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5-NGG-3).
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 (88.4).
3. Scans reference genome for off-target sites; validates 0 off-targets with CFD > 0.02.
4. Thesis Proposer and Antithesis Challenger cross-examine architectural edge cases.
5. Synthesis Agent crafts compromise; Judge verifies mathematical soundness and issues ruling.

- **Sovereign Outcome**: Optimal sgRNA candidate exported for sickle-cell gene editing. Dialectical consensus ratified by Judge; all hallucinations rigorously eliminated.

Domain 751–800: Microsoft 365 Copilot, Substrate & Declarative Agent Extensibility

Scenario 751: Substrate Semantic Index Item Ingestion with ACLs (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_substrate\_ingest, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash
```

```
tgs copilot substrate --connection-id tagisan_repo --item-id spec-auth-01 --title 'Auth Architecture' --content 'Architecture Decision Record for Zero-Trust Auth'
```

```
```
```

- **Execution Flow**:

1. Formats JSON schema for Microsoft Substrate external item.
2. Applies Entra ID tenant grant ACLs for user and guest principals.
3. Pushes external document to `/external/connections/tagisan_repo/items/spec-auth-01`.
4. Indexes document into Microsoft Copilot Semantic Index.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Architecture document indexed into Substrate; retrievable via M365 Copilot BizChat with 0ms sync delay.

Scenario 752: Declarative Agent v1.17 Manifest Generation (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_declarative\_agent, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot agent --name 'CodebaseArchitect' --schema 1.17 --instructions 'Analyze Rust AST and enforce 1ES SDL security standards'
```

```
```
```

- **Execution Flow**:

1. Generates declarative agent JSON schema conforming to v1.17 specification.
 2. Configures instruction prompts and capabilities definitions.
 3. Grounds agent with Graph Connectors and repository search indexes.
 4. Validates schema against Microsoft 365 Copilot agent manifest validator.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Ready-to-deploy declarative agent package generated for Microsoft 365 Copilot side-loading.

Scenario 753: Copilot Studio OpenAPI 3.0 Packager & Sideloadable Bundle (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_studio\_packager, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot studio --output dist/tgs_copilot_plugin.zip --server-url https://api.tagisan.internal
```

```
```
```

- **Execution Flow**:

1. Generates valid OpenAPI 3.0.3 specification exposing all 43 autonomous tools.
 2. Creates ai-plugin.json with authentication and operation descriptors.
 3. Generates Microsoft Teams app manifest.json and color/outline PNG icons.
 4. Assembles all files into an in-memory PKZIP archive bundle.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Production PKZIP package generated; 1-click importable into Microsoft Copilot Studio and Power Platform.

Scenario 754: Windows Web Account Manager (WAM) Silent PRT SSO (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_wam\_auth, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot wam --account engineer@corp.microsoft.com --scope 'https://graph.microsoft.com/.default'

```
```

- **Execution Flow**:

1. Invokes Windows WAM native broker protocol interface.
 2. Extracts device Primary Refresh Token (PRT) from TPM secure enclave.
 3. Acquires access token silently without user interaction or browser popups.
 4. Injects token into Tagisan HTTP client session store.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Zero-touch corporate SSO completed; valid Graph token acquired with zero user friction.

Scenario 755: Continuous Access Evaluation (CAE) Claims Step-Up Challenge (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_cae\_handler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot auth --handle-cae --claims-challenge
'eyJhY2Nlc3NfdG9rZW4iOmsibmJmIjpw7ImVzc2VudGlhbCI6dHJ1ZX19fQ=='

```
```

- **Execution Flow**:

1. Detects 401 Unauthorized with WWW-Authenticate: Bearer error='insufficient_claims'.
 2. Parses Base64-encoded JSON claims challenge from HTTP response headers.
 3. Prompts Windows Hello for Business biometric or FIDO2 key re-attestation.
 4. Re-executes token acquisition with claims parameter attached.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Conditional access step-up verified; downstream Graph request replayed and succeeded.

Scenario 756: Outlook Calendar Meeting Technical Pre-Read Briefing (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_calendar\_briefing, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot calendar --event-id 'AAMkAGI2AAAU=' --generate-preread

```
```

- **Execution Flow**:
 1. Queries Microsoft Graph API for meeting attendees, subject, and attachments.
 2. Analyzes attendee roles from Entra ID organization hierarchy.
 3. Correlates meeting topic with relevant Git pull requests and architecture specs.
 4. Synthesizes 2-page executive technical briefing with action item checklists.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Pre-meeting briefing card generated and broadcast to Outlook attendees before call starts.

Scenario 757: Teams Meeting Audio Transcript Action Item Extraction (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_teams\_meeting, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot teams --transcript-file meeting_0916.vtt --extract-actions

```
```

- **Execution Flow**:
 1. Parses WebVTT / JSON transcript stream with speaker diarization.
 2. Filters conversational filler and isolates architectural consensus statements.
 3. Identifies assigned action items, responsible owners, and target deadlines.
 4. Cross-checks extracted tasks against active Azure DevOps sprint backlog.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Structured action items synthesized; batch-imported into ADO sprint with assignees linked.

Scenario 758: Streaming Quantitative Models into Excel via Office.js Custom Functions (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_excel\_addin`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot excel --model black_scholes --output dist/tgs_excel_addin.zip

```
```

- **Execution Flow**:

1. Compiles Rust mathematical formulas into JavaScript WebAssembly modules.
2. Generates Office.js Custom Functions JSON metadata and manifest.xml.
3. Configures real-time streaming WebSocket endpoint for live ticker feeds.
4. Validates Excel add-in manifest against Office Store ingestion schema.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Excel add-in packaged; custom function =TAGISAN.PRICING() live in desktop and web Excel.

Scenario 759: Executive Presentation Deck Synthesis in OpenXML PowerPoint (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_presentation`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot pptx --title 'Q3 Sovereign Agent Architecture' --slides 12 --output deck.pptx

```
```

- **Execution Flow**:

1. Builds OpenXML presentation package (.pptx) with modern slide layouts.
2. Embeds architectural diagrams, KPI metric cards, and timeline milestones.
3. Applies corporate brand color palette and typography styles.
4. Generates presenter speaker notes for every slide.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Professional 12-slide executive deck compiled in 1.2 seconds without external office dependencies.

Scenario 760: Enterprise SharePoint Online Library Knowledge Crawling (Iteration 1): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_sharepoint\_crawler, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot sharepoint --site-url 'https://corp.sharepoint.com/teams/CoreArch' --depth 3

```
```

- **Execution Flow**:

1. Authenticates against SharePoint Online via Graph API app credentials.
 2. Traverses document libraries, folders, site pages, and metadata lists.
 3. Extracts text from DOCX, PDF, XLSX, and PPTX binaries.
 4. Computes vector embeddings and indexes content into PILOT semantic memory.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Over 2,400 corporate documents indexed and grounded for conversational Copilot queries.

Scenario 761: Substrate Semantic Index Item Ingestion with ACLs (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_substrate\_ingest, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot substrate --connection-id tagisan_repo --item-id spec-auth-01 --title 'Auth Architecture' --content 'Architecture Decision Record for Zero-Trust Auth'

```
```

- **Execution Flow**:

1. Formats JSON schema for Microsoft Substrate external item.
 2. Applies Entra ID tenant grant ACLs for user and guest principals.
 3. Pushes external document to /external/connections/tagisan_repo/items/spec-auth-01.
 4. Indexes document into Microsoft Copilot Semantic Index.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Architecture document indexed into Substrate; retrievable via M365 Copilot BizChat with 0ms sync delay.

Scenario 762: Declarative Agent v1.17 Manifest Generation (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_declarative\_agent, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot agent --name 'CodebaseArchitect' --schema 1.17 --instructions 'Analyze Rust AST and enforce 1ES SDL security standards'
```

```
```
```

- **Execution Flow**:

1. Generates declarative agent JSON schema conforming to v1.17 specification.
 2. Configures instruction prompts and capabilities definitions.
 3. Grounds agent with Graph Connectors and repository search indexes.
 4. Validates schema against Microsoft 365 Copilot agent manifest validator.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Ready-to-deploy declarative agent package generated for Microsoft 365 Copilot side-loading.

Scenario 763: Copilot Studio OpenAPI 3.0 Packager & Sideloadable Bundle (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_studio\_packager, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot studio --output dist/tgs_copilot_plugin.zip --server-url https://api.tagisan.internal
```

```
```
```

- **Execution Flow**:

1. Generates valid OpenAPI 3.0.3 specification exposing all 43 autonomous tools.
 2. Creates ai-plugin.json with authentication and operation descriptors.
 3. Generates Microsoft Teams app manifest.json and color/outline PNG icons.
 4. Assembles all files into an in-memory PKZIP archive bundle.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Production PKZIP package generated; 1-click importable into Microsoft Copilot Studio and Power Platform.

Scenario 764: Windows Web Account Manager (WAM) Silent PRT SSO (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_wam\_auth, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot wam --account engineer@corp.microsoft.com --scope 'https://graph.microsoft.com/.default'

```
```

- **Execution Flow**:

1. Invokes Windows WAM native broker protocol interface.
 2. Extracts device Primary Refresh Token (PRT) from TPM secure enclave.
 3. Acquires access token silently without user interaction or browser popups.
 4. Injects token into Tagisan HTTP client session store.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Zero-touch corporate SSO completed; valid Graph token acquired with zero user friction.

Scenario 765: Continuous Access Evaluation (CAE) Claims Step-Up Challenge (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_cae\_handler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot auth --handle-cae --claims-challenge
'eyJhY2Nlc3NfdG9rZW4iOmsibmJmIjpw7ImVzc2VudGlhbCI6dHJ1ZX19fQ=='

```
```

- **Execution Flow**:

1. Detects 401 Unauthorized with WWW-Authenticate: Bearer error='insufficient_claims'.
 2. Parses Base64-encoded JSON claims challenge from HTTP response headers.
 3. Prompts Windows Hello for Business biometric or FIDO2 key re-attestation.
 4. Re-executes token acquisition with claims parameter attached.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Conditional access step-up verified; downstream Graph request replayed and succeeded.

Scenario 766: Outlook Calendar Meeting Technical Pre-Read Briefing (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_calendar\_briefing, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot calendar --event-id 'AAMkAGI2AAAU=' --generate-preread

```
```

- **Execution Flow**:

1. Queries Microsoft Graph API for meeting attendees, subject, and attachments.
 2. Analyzes attendee roles from Entra ID organization hierarchy.
 3. Correlates meeting topic with relevant Git pull requests and architecture specs.
 4. Synthesizes 2-page executive technical briefing with action item checklists.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Pre-meeting briefing card generated and broadcast to Outlook attendees before call starts.

Scenario 767: Teams Meeting Audio Transcript Action Item Extraction (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_teams\_meeting, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot teams --transcript-file meeting_0916.vtt --extract-actions

```
```

- **Execution Flow**:

1. Parses WebVTT / JSON transcript stream with speaker diarization.
 2. Filters conversational filler and isolates architectural consensus statements.
 3. Identifies assigned action items, responsible owners, and target deadlines.
 4. Cross-checks extracted tasks against active Azure DevOps sprint backlog.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Structured action items synthesized; batch-imported into ADO sprint with assignees linked.

Scenario 768: Streaming Quantitative Models into Excel via Office.js Custom Functions (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_excel\_addin`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot excel --model black_scholes --output dist/tgs_excel_addin.zip

```
```

- **Execution Flow**:

1. Compiles Rust mathematical formulas into JavaScript WebAssembly modules.
2. Generates Office.js Custom Functions JSON metadata and manifest.xml.
3. Configures real-time streaming WebSocket endpoint for live ticker feeds.
4. Validates Excel add-in manifest against Office Store ingestion schema.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Excel add-in packaged; custom function =TAGISAN.PRICING() live in desktop and web Excel.

Scenario 769: Executive Presentation Deck Synthesis in OpenXML PowerPoint (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_presentation`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot pptx --title 'Q3 Sovereign Agent Architecture' --slides 12 --output deck.pptx

```
```

- **Execution Flow**:

1. Builds OpenXML presentation package (.pptx) with modern slide layouts.
2. Embeds architectural diagrams, KPI metric cards, and timeline milestones.
3. Applies corporate brand color palette and typography styles.
4. Generates presenter speaker notes for every slide.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Professional 12-slide executive deck compiled in 1.2 seconds without external office dependencies.

Scenario 770: Enterprise SharePoint Online Library Knowledge Crawling (Iteration 2): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_sharepoint\_crawler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot sharepoint --site-url 'https://corp.sharepoint.com/teams/CoreArch' --depth 3
```

```
```
```

- **Execution Flow**:

1. Authenticates against SharePoint Online via Graph API app credentials.
 2. Traverses document libraries, folders, site pages, and metadata lists.
 3. Extracts text from DOCX, PDF, XLSX, and PPTX binaries.
 4. Computes vector embeddings and indexes content into PILOT semantic memory.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Over 2,400 corporate documents indexed and grounded for conversational Copilot queries.

Scenario 771: Substrate Semantic Index Item Ingestion with ACLs (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_substrate\_ingest, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot substrate --connection-id tagisan_repo --item-id spec-auth-01 --title 'Auth Architecture' --content 'Architecture Decision Record for Zero-Trust Auth'
```

```
```
```

- **Execution Flow**:

1. Formats JSON schema for Microsoft Substrate external item.
 2. Applies Entra ID tenant grant ACLs for user and guest principals.
 3. Pushes external document to /external/connections/tagisan_repo/items/spec-auth-01.
 4. Indexes document into Microsoft Copilot Semantic Index.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Architecture document indexed into Substrate; retrievable via M365 Copilot BizChat with 0ms sync delay.

Scenario 772: Declarative Agent v1.17 Manifest Generation (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_declarative\_agent, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot agent --name 'CodebaseArchitect' --schema 1.17 --instructions 'Analyze Rust AST and enforce 1ES SDL security standards'
```

```
```
```

- **Execution Flow**:

1. Generates declarative agent JSON schema conforming to v1.17 specification.
 2. Configures instruction prompts and capabilities definitions.
 3. Grounds agent with Graph Connectors and repository search indexes.
 4. Validates schema against Microsoft 365 Copilot agent manifest validator.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Ready-to-deploy declarative agent package generated for Microsoft 365 Copilot side-loading.

Scenario 773: Copilot Studio OpenAPI 3.0 Packager & Sideloadable Bundle (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_studio\_packager, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot studio --output dist/tgs_copilot_plugin.zip --server-url https://api.tagisan.internal
```

```
```
```

- **Execution Flow**:

1. Generates valid OpenAPI 3.0.3 specification exposing all 43 autonomous tools.
 2. Creates ai-plugin.json with authentication and operation descriptors.
 3. Generates Microsoft Teams app manifest.json and color/outline PNG icons.
 4. Assembles all files into an in-memory PKZIP archive bundle.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Production PKZIP package generated; 1-click importable into Microsoft Copilot Studio and Power Platform.

Scenario 774: Windows Web Account Manager (WAM) Silent PRT SSO (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_wam\_auth, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot wam --account engineer@corp.microsoft.com --scope 'https://graph.microsoft.com/.default'

```
```

- **Execution Flow**:

1. Invokes Windows WAM native broker protocol interface.
 2. Extracts device Primary Refresh Token (PRT) from TPM secure enclave.
 3. Acquires access token silently without user interaction or browser popups.
 4. Injects token into Tagisan HTTP client session store.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Zero-touch corporate SSO completed; valid Graph token acquired with zero user friction.

Scenario 775: Continuous Access Evaluation (CAE) Claims Step-Up Challenge (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_cae\_handler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot auth --handle-cae --claims-challenge
'eyJhY2Nlc3NfdG9rZW4iOmsibmJmIjpw7ImVzc2VudGlhbCI6dHJ1ZX19fQ=='

```
```

- **Execution Flow**:

1. Detects 401 Unauthorized with WWW-Authenticate: Bearer error='insufficient_claims'.
 2. Parses Base64-encoded JSON claims challenge from HTTP response headers.
 3. Prompts Windows Hello for Business biometric or FIDO2 key re-attestation.
 4. Re-executes token acquisition with claims parameter attached.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Conditional access step-up verified; downstream Graph request replayed and succeeded.

Scenario 776: Outlook Calendar Meeting Technical Pre-Read Briefing (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_calendar\_briefing, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot calendar --event-id 'AAMkAGI2AAAU=' --generate-preread

```
```

- **Execution Flow**:

1. Queries Microsoft Graph API for meeting attendees, subject, and attachments.
 2. Analyzes attendee roles from Entra ID organization hierarchy.
 3. Correlates meeting topic with relevant Git pull requests and architecture specs.
 4. Synthesizes 2-page executive technical briefing with action item checklists.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Pre-meeting briefing card generated and broadcast to Outlook attendees before call starts.

Scenario 777: Teams Meeting Audio Transcript Action Item Extraction (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_teams\_meeting, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot teams --transcript-file meeting_0916.vtt --extract-actions

```
```

- **Execution Flow**:

1. Parses WebVTT / JSON transcript stream with speaker diarization.
 2. Filters conversational filler and isolates architectural consensus statements.
 3. Identifies assigned action items, responsible owners, and target deadlines.
 4. Cross-checks extracted tasks against active Azure DevOps sprint backlog.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Structured action items synthesized; batch-imported into ADO sprint with assignees linked.

Scenario 778: Streaming Quantitative Models into Excel via Office.js Custom Functions (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_excel\_addin`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot excel --model black_scholes --output dist/tgs_excel_addin.zip

```
```

- **Execution Flow**:

1. Compiles Rust mathematical formulas into JavaScript WebAssembly modules.
2. Generates Office.js Custom Functions JSON metadata and manifest.xml.
3. Configures real-time streaming WebSocket endpoint for live ticker feeds.
4. Validates Excel add-in manifest against Office Store ingestion schema.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Excel add-in packaged; custom function =TAGISAN.PRICING() live in desktop and web Excel.

Scenario 779: Executive Presentation Deck Synthesis in OpenXML PowerPoint (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_presentation`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot pptx --title 'Q3 Sovereign Agent Architecture' --slides 12 --output deck.pptx

```
```

- **Execution Flow**:

1. Builds OpenXML presentation package (.pptx) with modern slide layouts.
2. Embeds architectural diagrams, KPI metric cards, and timeline milestones.
3. Applies corporate brand color palette and typography styles.
4. Generates presenter speaker notes for every slide.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Professional 12-slide executive deck compiled in 1.2 seconds without external office dependencies.

Scenario 780: Enterprise SharePoint Online Library Knowledge Crawling (Iteration 3): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_sharepoint\_crawler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot sharepoint --site-url 'https://corp.sharepoint.com/teams/CoreArch' --depth 3
```

```
```
```

- **Execution Flow**:

1. Authenticates against SharePoint Online via Graph API app credentials.
 2. Traverses document libraries, folders, site pages, and metadata lists.
 3. Extracts text from DOCX, PDF, XLSX, and PPTX binaries.
 4. Computes vector embeddings and indexes content into PILOT semantic memory.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Over 2,400 corporate documents indexed and grounded for conversational Copilot queries.

Scenario 781: Substrate Semantic Index Item Ingestion with ACLs (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_substrate\_ingest, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot substrate --connection-id tagisan_repo --item-id spec-auth-01 --title 'Auth Architecture' --content 'Architecture Decision Record for Zero-Trust Auth'
```

```
```
```

- **Execution Flow**:

1. Formats JSON schema for Microsoft Substrate external item.
 2. Applies Entra ID tenant grant ACLs for user and guest principals.
 3. Pushes external document to /external/connections/tagisan_repo/items/spec-auth-01.
 4. Indexes document into Microsoft Copilot Semantic Index.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Architecture document indexed into Substrate; retrievable via M365 Copilot BizChat with 0ms sync delay.

Scenario 782: Declarative Agent v1.17 Manifest Generation (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_declarative\_agent, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot agent --name 'CodebaseArchitect' --schema 1.17 --instructions 'Analyze Rust AST and enforce 1ES SDL security standards'
```

```
```
```

- **Execution Flow**:

1. Generates declarative agent JSON schema conforming to v1.17 specification.
 2. Configures instruction prompts and capabilities definitions.
 3. Grounds agent with Graph Connectors and repository search indexes.
 4. Validates schema against Microsoft 365 Copilot agent manifest validator.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Ready-to-deploy declarative agent package generated for Microsoft 365 Copilot side-loading.

Scenario 783: Copilot Studio OpenAPI 3.0 Packager & Sideloadable Bundle (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_studio\_packager, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot studio --output dist/tgs_copilot_plugin.zip --server-url https://api.tagisan.internal
```

```
```
```

- **Execution Flow**:

1. Generates valid OpenAPI 3.0.3 specification exposing all 43 autonomous tools.
 2. Creates ai-plugin.json with authentication and operation descriptors.
 3. Generates Microsoft Teams app manifest.json and color/outline PNG icons.
 4. Assembles all files into an in-memory PKZIP archive bundle.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Production PKZIP package generated; 1-click importable into Microsoft Copilot Studio and Power Platform.

Scenario 784: Windows Web Account Manager (WAM) Silent PRT SSO (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_wam\_auth, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot wam --account engineer@corp.microsoft.com --scope 'https://graph.microsoft.com/.default'

```
```

- **Execution Flow**:

1. Invokes Windows WAM native broker protocol interface.
 2. Extracts device Primary Refresh Token (PRT) from TPM secure enclave.
 3. Acquires access token silently without user interaction or browser popups.
 4. Injects token into Tagisan HTTP client session store.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Zero-touch corporate SSO completed; valid Graph token acquired with zero user friction.

Scenario 785: Continuous Access Evaluation (CAE) Claims Step-Up Challenge (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_cae\_handler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot auth --handle-cae --claims-challenge
'eyJhY2Nlc3NfdG9rZW4iOmsibmJmIjp7ImVzc2VudGlhbCI6dHJ1ZX19fQ=='

```
```

- **Execution Flow**:

1. Detects 401 Unauthorized with WWW-Authenticate: Bearer error='insufficient_claims'.
 2. Parses Base64-encoded JSON claims challenge from HTTP response headers.
 3. Prompts Windows Hello for Business biometric or FIDO2 key re-attestation.
 4. Re-executes token acquisition with claims parameter attached.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Conditional access step-up verified; downstream Graph request replayed and succeeded.

Scenario 786: Outlook Calendar Meeting Technical Pre-Read Briefing (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_calendar\_briefing, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot calendar --event-id 'AAMkAGI2AAAU=' --generate-preread

```
```

- **Execution Flow**:

1. Queries Microsoft Graph API for meeting attendees, subject, and attachments.
 2. Analyzes attendee roles from Entra ID organization hierarchy.
 3. Correlates meeting topic with relevant Git pull requests and architecture specs.
 4. Synthesizes 2-page executive technical briefing with action item checklists.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Pre-meeting briefing card generated and broadcast to Outlook attendees before call starts.

Scenario 787: Teams Meeting Audio Transcript Action Item Extraction (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_teams\_meeting, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot teams --transcript-file meeting_0916.vtt --extract-actions

```
```

- **Execution Flow**:

1. Parses WebVTT / JSON transcript stream with speaker diarization.
 2. Filters conversational filler and isolates architectural consensus statements.
 3. Identifies assigned action items, responsible owners, and target deadlines.
 4. Cross-checks extracted tasks against active Azure DevOps sprint backlog.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Structured action items synthesized; batch-imported into ADO sprint with assignees linked.

Scenario 788: Streaming Quantitative Models into Excel via Office.js Custom Functions (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_excel\_addin`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot excel --model black_scholes --output dist/tgs_excel_addin.zip

```
```

- **Execution Flow**:

1. Compiles Rust mathematical formulas into JavaScript WebAssembly modules.
2. Generates Office.js Custom Functions JSON metadata and manifest.xml.
3. Configures real-time streaming WebSocket endpoint for live ticker feeds.
4. Validates Excel add-in manifest against Office Store ingestion schema.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Excel add-in packaged; custom function =TAGISAN.PRICING() live in desktop and web Excel.

Scenario 789: Executive Presentation Deck Synthesis in OpenXML PowerPoint (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_presentation`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot pptx --title 'Q3 Sovereign Agent Architecture' --slides 12 --output deck.pptx

```
```

- **Execution Flow**:

1. Builds OpenXML presentation package (.pptx) with modern slide layouts.
2. Embeds architectural diagrams, KPI metric cards, and timeline milestones.
3. Applies corporate brand color palette and typography styles.
4. Generates presenter speaker notes for every slide.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Professional 12-slide executive deck compiled in 1.2 seconds without external office dependencies.

Scenario 790: Enterprise SharePoint Online Library Knowledge Crawling (Iteration 4): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_sharepoint\_crawler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot sharepoint --site-url 'https://corp.sharepoint.com/teams/CoreArch' --depth 3

```
```

- **Execution Flow**:

1. Authenticates against SharePoint Online via Graph API app credentials.
 2. Traverses document libraries, folders, site pages, and metadata lists.
 3. Extracts text from DOCX, PDF, XLSX, and PPTX binaries.
 4. Computes vector embeddings and indexes content into PILOT semantic memory.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Over 2,400 corporate documents indexed and grounded for conversational Copilot queries.

Scenario 791: Substrate Semantic Index Item Ingestion with ACLs (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_substrate\_ingest, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot substrate --connection-id tagisan_repo --item-id spec-auth-01 --title 'Auth Architecture' --content 'Architecture Decision Record for Zero-Trust Auth'

```
```

- **Execution Flow**:

1. Formats JSON schema for Microsoft Substrate external item.
 2. Applies Entra ID tenant grant ACLs for user and guest principals.
 3. Pushes external document to /external/connections/tagisan_repo/items/spec-auth-01.
 4. Indexes document into Microsoft Copilot Semantic Index.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Architecture document indexed into Substrate; retrievable via M365 Copilot BizChat with 0ms sync delay.

Scenario 792: Declarative Agent v1.17 Manifest Generation (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_declarative\_agent, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot agent --name 'CodebaseArchitect' --schema 1.17 --instructions 'Analyze Rust AST and enforce 1ES SDL security standards'
```

```
```
```

- **Execution Flow**:

1. Generates declarative agent JSON schema conforming to v1.17 specification.
 2. Configures instruction prompts and capabilities definitions.
 3. Grounds agent with Graph Connectors and repository search indexes.
 4. Validates schema against Microsoft 365 Copilot agent manifest validator.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Ready-to-deploy declarative agent package generated for Microsoft 365 Copilot side-loading.

Scenario 793: Copilot Studio OpenAPI 3.0 Packager & Sideloadable Bundle (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_studio\_packager, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash
```

```
tgs copilot studio --output dist/tgs_copilot_plugin.zip --server-url https://api.tagisan.internal
```

```
```
```

- **Execution Flow**:

1. Generates valid OpenAPI 3.0.3 specification exposing all 43 autonomous tools.
 2. Creates ai-plugin.json with authentication and operation descriptors.
 3. Generates Microsoft Teams app manifest.json and color/outline PNG icons.
 4. Assembles all files into an in-memory PKZIP archive bundle.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Production PKZIP package generated; 1-click importable into Microsoft Copilot Studio and Power Platform.

Scenario 794: Windows Web Account Manager (WAM) Silent PRT SSO (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_wam\_auth, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot wam --account engineer@corp.microsoft.com --scope 'https://graph.microsoft.com/.default'

```
```

- **Execution Flow**:

1. Invokes Windows WAM native broker protocol interface.
 2. Extracts device Primary Refresh Token (PRT) from TPM secure enclave.
 3. Acquires access token silently without user interaction or browser popups.
 4. Injects token into Tagisan HTTP client session store.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Zero-touch corporate SSO completed; valid Graph token acquired with zero user friction.

Scenario 795: Continuous Access Evaluation (CAE) Claims Step-Up Challenge (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_cae\_handler, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot auth --handle-cae --claims-challenge
'eyJhY2Nlc3NfdG9rZW4iOmsibmJmIjpw7ImVzc2VudGlhbCI6dHJ1ZX19fQ=='

```
```

- **Execution Flow**:

1. Detects 401 Unauthorized with WWW-Authenticate: Bearer error='insufficient_claims'.
 2. Parses Base64-encoded JSON claims challenge from HTTP response headers.
 3. Prompts Windows Hello for Business biometric or FIDO2 key re-attestation.
 4. Re-executes token acquisition with claims parameter attached.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Conditional access step-up verified; downstream Graph request replayed and succeeded.

Scenario 796: Outlook Calendar Meeting Technical Pre-Read Briefing (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_calendar\_briefing, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot calendar --event-id 'AAMkAGI2AAAU=' --generate-preread

```
```

- **Execution Flow**:

1. Queries Microsoft Graph API for meeting attendees, subject, and attachments.
 2. Analyzes attendee roles from Entra ID organization hierarchy.
 3. Correlates meeting topic with relevant Git pull requests and architecture specs.
 4. Synthesizes 2-page executive technical briefing with action item checklists.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Pre-meeting briefing card generated and broadcast to Outlook attendees before call starts.

Scenario 797: Teams Meeting Audio Transcript Action Item Extraction (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_teams\_meeting, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot teams --transcript-file meeting_0916.vtt --extract-actions

```
```

- **Execution Flow**:

1. Parses WebVTT / JSON transcript stream with speaker diarization.
 2. Filters conversational filler and isolates architectural consensus statements.
 3. Identifies assigned action items, responsible owners, and target deadlines.
 4. Cross-checks extracted tasks against active Azure DevOps sprint backlog.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
- **Sovereign Outcome**: Structured action items synthesized; batch-imported into ADO sprint with assignees linked.

Scenario 798: Streaming Quantitative Models into Excel via Office.js Custom Functions (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_excel\_addin`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot excel --model black_scholes --output dist/tgs_excel_addin.zip

```
```

- **Execution Flow**:

1. Compiles Rust mathematical formulas into JavaScript WebAssembly modules.
2. Generates Office.js Custom Functions JSON metadata and manifest.xml.
3. Configures real-time streaming WebSocket endpoint for live ticker feeds.
4. Validates Excel add-in manifest against Office Store ingestion schema.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Excel add-in packaged; custom function =TAGISAN.PRICING() live in desktop and web Excel.

Scenario 799: Executive Presentation Deck Synthesis in OpenXML PowerPoint (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_presentation`, Microsoft 365 Copilot, Graph API, Entra ID`

- **Command**:

```
```bash

tgs copilot pptx --title 'Q3 Sovereign Agent Architecture' --slides 12 --output deck.pptx

```
```

- **Execution Flow**:

1. Builds OpenXML presentation package (.pptx) with modern slide layouts.
2. Embeds architectural diagrams, KPI metric cards, and timeline milestones.
3. Applies corporate brand color palette and typography styles.
4. Generates presenter speaker notes for every slide.
5. Validates cryptographic signatures and confirms telemetry persistence in audit log.

- **Sovereign Outcome**: Professional 12-slide executive deck compiled in 1.2 seconds without external office dependencies.

Scenario 800: Enterprise SharePoint Online Library Knowledge Crawling (Iteration 5): Microsoft 365 Copilot Enterprise

- **TGS Capabilities**: `copilot\_sharepoint\_crawler, Microsoft 365 Copilot, Graph API, Entra ID`
- **Command**:

```
```bash

tgs copilot sharepoint --site-url 'https://corp.sharepoint.com/teams/CoreArch' --depth 3

```
```

- **Execution Flow**:
 1. Authenticates against SharePoint Online via Graph API app credentials.
 2. Traverses document libraries, folders, site pages, and metadata lists.
 3. Extracts text from DOCX, PDF, XLSX, and PPTX binaries.
 4. Computes vector embeddings and indexes content into PILOT semantic memory.
 5. Validates cryptographic signatures and confirms telemetry persistence in audit log.
 - **Sovereign Outcome**: Over 2,400 corporate documents indexed and grounded for conversational Copilot queries.

Domain 801–850: ⚡ Microsoft Power Platform, Dataverse, Power Automate & Power BI

Scenario 801: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 1): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
 - **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 802: Power Apps Self-Service Codebase Release Gate Portal (Iteration 1): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:

1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 803: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 1): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:

1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 804: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 1): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 805: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 1): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 806: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 2): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 807: Power Apps Self-Service Codebase Release Gate Portal (Iteration 2): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 808: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 2): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:

1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 809: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 2): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 810: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 2): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json
```

```
```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 811: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 3): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'
```

```
```
```

- **Execution Flow**:

1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 812: Power Apps Self-Service Codebase Release Gate Portal (Iteration 3): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 813: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 3): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 814: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 3): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 815: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 3): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 816: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 4): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 817: Power Apps Self-Service Codebase Release Gate Portal (Iteration 4): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 818: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 4): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:

1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 819: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 4): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 820: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 4): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json
```

```
```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 821: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 5): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'
```

```
```
```

- **Execution Flow**:

1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 822: Power Apps Self-Service Codebase Release Gate Portal (Iteration 5): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 823: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 5): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 824: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 5): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 825: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 5): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 826: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 6): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 827: Power Apps Self-Service Codebase Release Gate Portal (Iteration 6): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 828: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 6): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 829: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 6): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:
 1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 830: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 6): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json
```

```
```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 831: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 7): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'
```

```
```
```

- **Execution Flow**:

1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 832: Power Apps Self-Service Codebase Release Gate Portal (Iteration 7): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 833: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 7): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 834: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 7): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 835: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 7): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 836: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 8): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 837: Power Apps Self-Service Codebase Release Gate Portal (Iteration 8): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 838: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 8): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 839: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 8): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:
 1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 840: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 8): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json
```

```
```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 841: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 9): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'
```

```
```
```

- **Execution Flow**:

1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 842: Power Apps Self-Service Codebase Release Gate Portal (Iteration 9): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 843: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 9): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json

```
```

- **Execution Flow**:
 1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
 2. Maps multi-agent debate transcripts and security decisions to CDM schema.
 3. Executes batch upsert operation using OData single-transaction changeset.
 4. Validates row-level access permissions and column auditing flags.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 844: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 9): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics

```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
 2. Formats telemetry into Power BI REST API streaming dataset schema.
 3. Pushes telemetry batches to Power BI service via HTTPS POST.
 4. Triggers real-time dashboard visual refresh across engineering workspaces.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 845: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 9): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:

1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.
- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Scenario 846: Power Automate Cloud Flow Triggered AST Code Audit (Iteration 10): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --flow-id 'flow-audit-pr-01' --repo '/src/backend'

```
```

- **Execution Flow**:
 1. Receives webhook event from Power Automate Cloud Flow upon PR creation.
 2. Executes pre-execution AST security and blast-radius evaluation.
 3. Formats compliance findings into Adaptive Card v1.5 JSON payload.
 4. Returns structured HTTP 200 response with risk score to Power Automate.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Automated self-healing workflow triggered; PR approval card posted to Microsoft Teams channel.

Scenario 847: Power Apps Self-Service Codebase Release Gate Portal (Iteration 10): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_studio\_packager, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-apps --portal-mode --listen-port 8088

```
```

- **Execution Flow**:
 1. Initializes lightweight embedded REST bridge for Power Apps Canvas integration.
 2. Exposes endpoints for code scan trigger, branch status, and SBOM download.
 3. Authenticates Canvas app requests via Entra ID user bearer token.
 4. Streams live scan progress updates via Server-Sent Events (SSE).
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Non-technical Product Managers trigger 1ES compliance audits from mobile Power App.

Scenario 848: Dataverse Common Data Model (CDM) Audit Trail Synchronization (Iteration 10): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_dataverse, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot dataverse --table 'tgs_audit_log' --sync-file audit_transcripts.json
```

```
```
```

- **Execution Flow**:

1. Authenticates against Microsoft Dataverse Web API with OAuth2 bearer token.
2. Maps multi-agent debate transcripts and security decisions to CDM schema.
3. Executes batch upsert operation using OData single-transaction changeset.
4. Validates row-level access permissions and column auditing flags.
5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: All autonomous agent decisions immutably recorded in Dataverse for SOX compliance.

Scenario 849: Power BI Real-Time Streaming Telemetry Dataset Ingestion (Iteration 10): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_powerbi, Power Platform, Dataverse, Power Automate, Power BI`

- **Command**:

```
```bash
```

```
tgs copilot powerbi --dataset 'tgs_codebase_health' --push-metrics
```

```
```
```

- **Execution Flow**:

1. Collects live AST cyclomatic complexity, Shannon entropy, and test coverage.
2. Formats telemetry into Power BI REST API streaming dataset schema.
3. Pushes telemetry batches to Power BI service via HTTPS POST.
4. Triggers real-time dashboard visual refresh across engineering workspaces.
5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Executive telemetry dashboard displays live codebase health and security posture.

Scenario 850: Power Automate Automated Incident Triage & Hotfix Generation (Iteration 10): Microsoft Power Platform

- **TGS Capabilities**: `copilot\_power\_automate, Power Platform, Dataverse, Power Automate, Power BI`
- **Command**:

```
```bash

tgs copilot power-automate --action triage_incident --incident-data incident_sev1.json

```
```

- **Execution Flow**:
 1. Ingests incident alert payload forwarded from Power Automate ServiceNow connector.
 2. Runs blast-radius analysis to locate faulty commits and crashing functions.
 3. Generates targeted code patch inside Landlock isolated sandbox.
 4. Runs automated unit and integration tests to verify bug resolution.
 5. Confirms transaction atomicity and writes trace event to OpenTelemetry sink.

- **Sovereign Outcome**: Zero-touch hotfix generated and attached to Power Automate approval notification.

Domain 851–900: Microsoft 1ES Security, CredScan, PoliCheck, SPDX SBOM & IcM Incident Bridge

Scenario 851: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 1): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`
- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:
 1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 852: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 1): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 853: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 1): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
 2. Computes SHA-256 package checksums and license declarations.
 3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
 4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 854: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 1): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action query

```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
 2. Correlates failure timestamp with recent Git commit history and deployment logs.
 3. Identifies suspicious commit introducing regression via AST blast-radius query.
 4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 855: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 1): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 856: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 2): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action credscan --file src/storage.rs
```

```
```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
4. Redacts secrets in terminal output and emits line-accurate violation list.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 857: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 2): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 858: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 2): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
2. Computes SHA-256 package checksums and license declarations.
3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 859: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 2): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot icm --incident-id 4920192 --action query
```

```
```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
2. Correlates failure timestamp with recent Git commit history and deployment logs.
3. Identifies suspicious commit introducing regression via AST blast-radius query.
4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 860: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 2): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 861: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 3): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 862: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 3): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 863: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 3): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
 2. Computes SHA-256 package checksums and license declarations.
 3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
 4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 864: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 3): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action query

```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
 2. Correlates failure timestamp with recent Git commit history and deployment logs.
 3. Identifies suspicious commit introducing regression via AST blast-radius query.
 4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 865: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 3): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 866: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 4): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action credscan --file src/storage.rs
```

```
```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
4. Redacts secrets in terminal output and emits line-accurate violation list.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 867: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 4): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 868: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 4): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
2. Computes SHA-256 package checksums and license declarations.
3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 869: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 4): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot icm --incident-id 4920192 --action query
```

```
```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
2. Correlates failure timestamp with recent Git commit history and deployment logs.
3. Identifies suspicious commit introducing regression via AST blast-radius query.
4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 870: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 4): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 871: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 5): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 872: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 5): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 873: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 5): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
 2. Computes SHA-256 package checksums and license declarations.
 3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
 4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 874: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 5): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action query

```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
 2. Correlates failure timestamp with recent Git commit history and deployment logs.
 3. Identifies suspicious commit introducing regression via AST blast-radius query.
 4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 875: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 5): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 876: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 6): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action credscan --file src/storage.rs
```

```
```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
4. Redacts secrets in terminal output and emits line-accurate violation list.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 877: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 6): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 878: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 6): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
2. Computes SHA-256 package checksums and license declarations.
3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 879: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 6): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot icm --incident-id 4920192 --action query
```

```
```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
2. Correlates failure timestamp with recent Git commit history and deployment logs.
3. Identifies suspicious commit introducing regression via AST blast-radius query.
4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 880: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 6): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 881: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 7): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 882: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 7): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action polichack --file src/orchestrator.rs

```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 883: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 7): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action sbom --file Cargo.lock

```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
 2. Computes SHA-256 package checksums and license declarations.
 3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
 4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 884: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 7): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action query

```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
 2. Correlates failure timestamp with recent Git commit history and deployment logs.
 3. Identifies suspicious commit introducing regression via AST blast-radius query.
 4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 885: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 7): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 886: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 8): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action credscan --file src/storage.rs
```

```
```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
4. Redacts secrets in terminal output and emits line-accurate violation list.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 887: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 8): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 888: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 8): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
2. Computes SHA-256 package checksums and license declarations.
3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 889: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 8): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot icm --incident-id 4920192 --action query
```

```
```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
2. Correlates failure timestamp with recent Git commit history and deployment logs.
3. Identifies suspicious commit introducing regression via AST blast-radius query.
4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 890: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 8): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 891: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 9): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 892: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 9): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action polichack --file src/orchestrator.rs
```

```
```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 893: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 9): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
 2. Computes SHA-256 package checksums and license declarations.
 3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
 4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 894: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 9): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action query

```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
 2. Correlates failure timestamp with recent Git commit history and deployment logs.
 3. Identifies suspicious commit introducing regression via AST blast-radius query.
 4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 895: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 9): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Scenario 896: Pre-Commit CredScan Shannon Entropy Secret Detection (Iteration 10): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action credscan --file src/storage.rs

```
```

- **Execution Flow**:

1. Scans source code with regular expression rules for Azure Storage AccountKeys.
 2. Evaluates Shannon entropy on suspicious tokens with threshold $H \geq 4.6$.
 3. Detects unencrypted cryptographic private key headers and ADO PAT tokens.
 4. Redacts secrets in terminal output and emits line-accurate violation list.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: CredScan audit completed; exposed secrets blocked from commit with zero false negatives.

Scenario 897: 1ES PoliCheck Geopolitical & Inclusivity Scanning (Iteration 10): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot sdl --action polichack --file src/orchestrator.rs

```
```

- **Execution Flow**:

1. Scans all identifiers, comments, and strings for non-inclusive terminology.
 2. Flags terms such as 'whitelist', 'blacklist', 'master', and 'slave'.
 3. Provides recommended corporate replacements ('allowlist', 'denylist', 'primary', 'secondary').
 4. Verifies geopolitical naming conventions conform to Microsoft 1ES guidelines.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: PoliCheck compliance passed; clean modern terminology established across codebase.

Scenario 898: SPDX 2.3 Software Bill of Materials (SBOM) Generation (Iteration 10): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_sdl\_audit`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot sdl --action sbom --file Cargo.lock
```

```
```
```

- **Execution Flow**:

1. Parses project dependency tree and package manifests from Cargo.lock.
2. Computes SHA-256 package checksums and license declarations.
3. Structures output in ISO/IEC 5962:2021 SPDX 2.3 JSON specification.
4. Embeds document namespace, creator tool identifiers, and timestamp metadata.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Enterprise-compliant SPDX 2.3 SBOM produced; validated for 1ES security gate merge.

Scenario 899: Microsoft IcM Live-Site Incident Diff Correlation (Iteration 10): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge`, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash
```

```
tgs copilot icm --incident-id 4920192 --action query
```

```
```
```

- **Execution Flow**:

1. Connects to Microsoft IcM REST API and retrieves Sev-1 ticket metadata.
2. Correlates failure timestamp with recent Git commit history and deployment logs.
3. Identifies suspicious commit introducing regression via AST blast-radius query.
4. Generates root-cause hypothesis backed by code diff and call-graph analysis.
5. Enforces pipeline gate check and updates compliance status in central repository.

- **Sovereign Outcome**: Incident root cause accurately pinned to specific commit within 1.8 seconds.

Scenario 900: Automated 5-Whys Post-Incident Review (PIR) Synthesis (Iteration 10): Microsoft 1ES Security & IcM

- **TGS Capabilities**: `copilot\_icm\_bridge, 1ES SDL, CredScan, PoliCheck, IcM, SPDX SBOM`

- **Command**:

```
```bash

tgs copilot icm --incident-id 4920192 --action draft_pir

```
```

- **Execution Flow**:

1. Extracts telemetry outage timeline, affected service endpoints, and user blast radius.
 2. Executes 5-Whys iterative causal analysis to reach root systemic vulnerability.
 3. Drafts comprehensive PIR markdown report with prevention action items.
 4. Generates Microsoft Teams Incident Bridge Adaptive Card v1.5 for war room review.
 5. Enforces pipeline gate check and updates compliance status in central repository.
- **Sovereign Outcome**: Executive-ready Post-Incident Review synthesized and dispatched to engineering leadership.

Domain 901–950: Frontier Microsoft Ecosystem: Visio 365, Access 365, Bicep, Defender & VS Code

Scenario 901: Visio 365 Architecture Vector Topology Synthesis (Iteration 1): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd

```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
 2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
 3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
 4. Validates vsdx package container integrity and ZIP compression structure.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 902: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 1): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot access --db-path inventory.accdb --export-schema sqlite

```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
 2. Extracts table schemas, primary keys, foreign relations, and indexes.
 3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
 4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 903: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 1): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep

```
```

- **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 904: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 1): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot defender --alert-id 'DEF-89210' --action remediate

```
```

- **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 905: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 1): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 906: Visio 365 Architecture Vector Topology Synthesis (Iteration 2): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd
```

```
```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
4. Validates vsdx package container integrity and ZIP compression structure.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 907: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 2): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot access --db-path inventory.accdb --export-schema sqlite
```

```
```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
2. Extracts table schemas, primary keys, foreign relations, and indexes.
3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 908: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 2): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep
```

```
```
```

• **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 909: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 2): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot defender --alert-id 'DEF-89210' --action remediate
```

```
```
```

• **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 910: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 2): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash  

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 911: Visio 365 Architecture Vector Topology Synthesis (Iteration 3): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash  

tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd

```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
 2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
 3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
 4. Validates vsdx package container integrity and ZIP compression structure.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 912: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 3): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot access --db-path inventory.accdb --export-schema sqlite

```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
 2. Extracts table schemas, primary keys, foreign relations, and indexes.
 3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
 4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 913: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 3): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep

```
```

- **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 914: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 3): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot defender --alert-id 'DEF-89210' --action remediate

```
```

- **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 915: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 3): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 916: Visio 365 Architecture Vector Topology Synthesis (Iteration 4): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd
```

```
```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
4. Validates vsdx package container integrity and ZIP compression structure.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 917: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 4): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot access --db-path inventory.accdb --export-schema sqlite
```

```
```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
2. Extracts table schemas, primary keys, foreign relations, and indexes.
3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 918: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 4): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep
```

```
```
```

• **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 919: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 4): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot defender --alert-id 'DEF-89210' --action remediate
```

```
```
```

• **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 920: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 4): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 921: Visio 365 Architecture Vector Topology Synthesis (Iteration 5): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd

```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
 2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
 3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
 4. Validates vsdx package container integrity and ZIP compression structure.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 922: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 5): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot access --db-path inventory.accdb --export-schema sqlite

```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
 2. Extracts table schemas, primary keys, foreign relations, and indexes.
 3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
 4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 923: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 5): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep

```
```

- **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 924: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 5): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot defender --alert-id 'DEF-89210' --action remediate

```
```

- **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 925: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 5): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 926: Visio 365 Architecture Vector Topology Synthesis (Iteration 6): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd
```

```
```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
4. Validates vsdx package container integrity and ZIP compression structure.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 927: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 6): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot access --db-path inventory.accdb --export-schema sqlite
```

```
```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
2. Extracts table schemas, primary keys, foreign relations, and indexes.
3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 928: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 6): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep
```

```
```
```

• **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 929: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 6): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot defender --alert-id 'DEF-89210' --action remediate
```

```
```
```

• **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 930: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 6): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 931: Visio 365 Architecture Vector Topology Synthesis (Iteration 7): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsdx

```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
 2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
 3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
 4. Validates vsdx package container integrity and ZIP compression structure.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 932: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 7): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot access --db-path inventory.accdb --export-schema sqlite

```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
 2. Extracts table schemas, primary keys, foreign relations, and indexes.
 3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
 4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 933: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 7): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep

```
```

- **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 934: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 7): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot defender --alert-id 'DEF-89210' --action remediate

```
```

- **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 935: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 7): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 936: Visio 365 Architecture Vector Topology Synthesis (Iteration 8): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd
```

```
```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
4. Validates vsdx package container integrity and ZIP compression structure.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 937: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 8): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot access --db-path inventory.accdb --export-schema sqlite
```

```
```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
2. Extracts table schemas, primary keys, foreign relations, and indexes.
3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 938: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 8): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep
```

```
```
```

• **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
3. Executes Bicep CLI linter and static validation check against Azure schema.
4. Emits parameters.json file with environment-specific overrides.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 939: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 8): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot defender --alert-id 'DEF-89210' --action remediate
```

```
```
```

• **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
2. Correlates alert telemetry with host file paths and binary hashes.
3. Detects unvalidated input parameter vulnerable to command injection.
4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 940: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 8): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash  

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 941: Visio 365 Architecture Vector Topology Synthesis (Iteration 9): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash  

tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd

```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
 2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
 3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
 4. Validates vsdx package container integrity and ZIP compression structure.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 942: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 9): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot access --db-path inventory.accdb --export-schema sqlite

```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
 2. Extracts table schemas, primary keys, foreign relations, and indexes.
 3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
 4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 943: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 9): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep

```
```

- **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
 2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
 3. Executes Bicep CLI linter and static validation check against Azure schema.
 4. Emits parameters.json file with environment-specific overrides.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 944: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 9): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot defender --alert-id 'DEF-89210' --action remediate

```
```

- **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
 2. Correlates alert telemetry with host file paths and binary hashes.
 3. Detects unvalidated input parameter vulnerable to command injection.
 4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 945: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 9): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash

tgs copilot vscode --lsp --port 9090

```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Scenario 946: Visio 365 Architecture Vector Topology Synthesis (Iteration 10): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_visio, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot visio --diagram-type architecture --input-graph topology.json --output arch.vsd
```

```
```
```

- **Execution Flow**:

1. Parses microservice component graph and network connectivity metadata.
2. Renders shapes, connectors, and boundaries using OpenXML Visio schema.
3. Applies Microsoft Azure / Cloud service stencils and auto-layout routing.
4. Validates vsdx package container integrity and ZIP compression structure.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Native Visio .vsdx architecture diagram synthesized and editable in Visio 365 Desktop and Web.

Scenario 947: Access 365 Legacy Database Schema Reverse-Engineering (Iteration 10): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_access, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot access --db-path inventory.accdb --export-schema sqlite
```

```
```
```

- **Execution Flow**:

1. Parses Jet/ACE binary database engine page headers and catalog tables.
2. Extracts table schemas, primary keys, foreign relations, and indexes.
3. Generates clean ANSI SQL DDL migration script for SQLite and PostgreSQL.
4. Flags deprecated VBA macro triggers and generates equivalent Rust handler stubs.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Legacy Access database reverse-engineered into modern relational schema in 420ms.

Scenario 948: Azure Bicep Infrastructure-as-Code Module Synthesis (Iteration 10): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_bicep, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot bicep --resource 'AzureKubernetesService' --tier 'Production' --output main.bicep
```

```
```
```

• **Execution Flow**:

1. Generates type-safe Bicep template conforming to Azure Resource Manager standards.
2. Enforces 1ES security policies (private endpoints, managed identities, TLS 1.3).
3. Executes Bicep CLI linter and static validation check against Azure schema.
4. Emits parameters.json file with environment-specific overrides.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Production-ready Bicep IaC module generated with zero security warnings.

Scenario 949: Microsoft Defender XDR Alert Ingestion & Remediation Patch (Iteration 10): Frontier Microsoft Ecosystem

• **TGS Capabilities**: `copilot\_defender, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

• **Command**:

```
```bash
```

```
tgs copilot defender --alert-id 'DEF-89210' --action remediate
```

```
```
```

• **Execution Flow**:

1. Queries Microsoft Defender XDR API for suspicious process execution alert.
2. Correlates alert telemetry with host file paths and binary hashes.
3. Detects unvalidated input parameter vulnerable to command injection.
4. Generates and tests code patch replacing shell invocation with sanitized `execvp()`.
5. Formats structured output and confirms compliance against Microsoft enterprise standards.

- **Sovereign Outcome**: Zero-day vulnerability patched and verified; Defender incident status updated to Resolved.

Scenario 950: VS Code LSP Extension Bridge & Inline CodeLens Blast Radius (Iteration 10): Frontier Microsoft Ecosystem

- **TGS Capabilities**: `copilot\_vscode, Visio 365, Access 365, Bicep, Defender, VS Code LSP`

- **Command**:

```
```bash
```

```
tgs copilot vscode --lsp --port 9090
```

```
```
```

- **Execution Flow**:

1. Launches Language Server Protocol (LSP) server communicating over JSON-RPC.
 2. Calculates real-time symbol blast radius on file save via CodeGraph AST.
 3. Provides CodeLens annotations displaying caller counts and downstream impact.
 4. Supplies quick-fix code actions for CredScan and PoliCheck violations.
 5. Formats structured output and confirms compliance against Microsoft enterprise standards.
- **Sovereign Outcome**: Seamless developer IDE experience enabled with instantaneous blast-radius feedback.

Module 13: Capstone Certification Labs & Practical Exams

Lab 1: Autonomous Self-Healing Code Pipeline

Objective: Configure Tagisan to monitor a Git repository, detect build errors, invoke a 3-round dialectical debate between a Refactoring Agent and a Security Auditor, apply the fixes, and verify tests pass.

- Prerequisites:** ``tgs run``, ``tgs debate``, GitHub repository clone.
- Verification:** Run ``cargo test`` inside the target project; all tests must pass.

Lab 2: Enterprise Market Intelligence Swarm

Objective: Build a multi-agent swarm orchestrating Brave Search MCP, PostgreSQL MCP, and Qdrant Vector Memory to monitor financial news filings and synthesize real-time market risk summaries.

- Prerequisites:** ``tgs mcp add brave-search-mcp``, ``tgs mcp add postgresql-mcp``, ``tgs mcp add qdrant-mcp``.
- Verification:** Query synthesized report via ``tgs ask`` verifying accurate RRF retrieval.

Lab 3: Air-Gapped Cyber Defense Cluster

Objective: Deploy Tagisan with zero outbound network connectivity (``TAGISAN_LOCAL_ONLY=1``), local Ollama Qwen2.5-Coder, and local SQLite memory. Verify AgentShield intercepts all simulated exfiltration attacks.

- Prerequisites:** ``ollama pull qwen2.5-coder:1.5b``, ``export TAGISAN_LOCAL_ONLY=1``.
- Verification:** Execute ``cargo test test_agentshield_sandbox``; all 8 security barriers must succeed.

Master Certification Checklist

- ☐ Build and verify native ``tgs`` release binary (``cargo build --release``).
- ☐ Connect Google Web OAuth (``tgs auth login gemini``) and test ``-m pro`` and ``-m lite``.
- ☐ Verify graceful offline fallback to local Ollama on logout (``tgs auth logout gemini``).
- ☐ Execute a multi-round Hegelian Dialectical Debate (``tgs debate``).
- ☐ Install and execute an external MCP tool from the 500 catalog.
- ☐ Implement a custom RFC-004 skill under ``.ecc/skills/``.
- ☐ Successfully execute at least 10 scenarios from Module 11 across different domains.
- ☐ Pass all 8 brutal security and transport tests in the test suite.